What we did since the call of 25 September

What to know before writing to Jérôme, then an explanation from scratch

Prepared for Giovanni Mantovani

14 October 2026

Abstract

The first two pages are what you need before replying. The rest explains the work from scratch and can be read in any order: **Part I** answers the eleven questions Jérôme asked on 25 September, **Part II** is the ϕ^4 field he and Marc proposed afterwards. Every number comes from a run registered in advance with its output frozen by a checksum (the few exceptions are flagged); a prediction that turned out wrong is reported in *italics*: a falsified prediction is a result, not a failure. **Part III** (Sections 6–15) is the mathematics and the technical details: derivations of everything used above, for reading with a pencil.

Before you write to Jérôme

What happened since the call. (1) His eleven questions were answered with experiments written down before they ran (table below). (2) The second laboratory he and Marc proposed, snapshots of the ϕ^4 (stochastic Allen–Cahn) field, was built and run: six settings, five estimators and an exact reference. (3) We tested where the method stops: a hidden variable shared by a whole sequence, subsampled chains, a chain outside the kernel family, data out of equilibrium. (4) BP was made cheaper (a coarser grid) and the "speciation" experiment he suggested was run. (5) On 12 October the *paper was rewritten* to follow the structure of their note of 30 September; the draft is waiting for your approval.

	Jérôme's question	Answer in one line
Q1	Gaussian messages = Gaussian score = Kalman smoother?	Yes, proved and checked to 2×10^{-13}. *Not new:* the thesis already had it; the paper just never said "Kalman". Forcing messages Gaussian on a grid fails numerically (error up to 0.26).
Q2	Is BP exact with a Laplace-like message family?	No: exact at the first site only; fails in the interior in 432 of 432 configurations.
Q3	What do the messages look like?	Very close to his guess (a Gaussian passed once through the kernel): KL 2,272× below a plain Gaussian's.
Q4	Can any finite family close the recursion?	The Gaussian does; his family does not; no proof for other families.
Q5	BP in Fourier space?	Not worth building: partly works as a representation, unreliably; our FFT solver is 5.8× *slower*.
Q6	Neural network as the kernel?	Loses on the paper's chain (64 of 64 comparisons) but wins where the simple family is wrong (below).
Q7	Time per sequence, network vs BP?	BP is slower: 12.5× the largest transformer on 8 threads, 3.2× on one thread. A coarser grid cuts BP by 4–6×.
Q8	Sample without diffusion?	Yes, exactly, and 731× cheaper than a 200-step reverse process.
Q9	Error at fixed t or pooled?	Per level: EM–BP beats every network at all 12 levels and the Gaussian score at the 8 ($M = 128$) or 10 ($M = 2048$) lowest.
Q10	Information curves: ρ, β or κ_2, κ_4?	Recomputed in cumulant coordinates; the message survives (fourth order falls about 700× more than second).
Q11	How does the cost scale in M and L?	Linearly in both; in the grid like $N_g^{1.2}$, not N_g^2 (the call's answer was wrong).

The four results that carry the story.

1. *The paper's chain.* A 25-number kernel learned from one noisy copy of each sequence, inside exact BP, beats every neural denoiser at every noise level and sample size, although the networks are trained on the clean sequences ($M = 2048$: error 0.023 against 0.060 for the best network).
2. *The ϕ^4 field.* The paper's own kernel (a linear drift) fails there: it does no better than the linear baseline. The field's own 5-number form, a Gaussian bond plus a quartic potential per site, is best on every setting: 3–17% of the linear baseline's error, and the best network is 14–24 times worse. Samples from it have the right kinks and domain lengths, and its fitted temperature is within 10% of the truth in 96 of 96 fits.
3. *The kernel family is the modelling choice.* A flexible (neural) kernel earns its parameters only when the simple family is wrong: on a chain with a sharp Laplace bond it is 1.8–2.5 times better, and on a chain subsampled every fourth site it is below the site energy in 16 of 16 seeds; where the simple family is right it loses.
4. *What exact inference costs.* BP is slower per sequence than a network, and the paper says so; the cheap lever is the grid (101 points suffice for the ϕ^4 kernels), not Fourier modes.

Where Jérôme may push, and what we did not show.

- The comparison with networks equalises the data, not the algorithm: the networks timed are the largest variants and in Part I they see clean sequences (they get more than EM–BP). The paper does not claim to rank learning algorithms.
- Several network-kernel results are upper bounds: the fit was still improving when its budget ended, so the true risk is lower.
- Speciation: commitment time behaves as predicted, but our prediction for the kink-energy law failed (fit 0.988 against $2\sqrt{2}/3 \approx 0.943$, within 2% predicted), so we do not claim the asymptotic constant; in one dimension there is no phase transition.
- A hidden variable shared by the whole sequence leaves a bias that no amount of data removes; augmenting the state repairs it.
- Some of our predictions were wrong and are reported as such (italics below). Three runs of 26 September were frozen after being read, which the registry says openly; an audit on 9 October re-derived the numbers and changed no verdict.

Waiting for you before you reply (details in Section 4): the framing (a general Markov kernel, the linear chain first and the field second, the field in the body) and the figure choice (the candidates are in Section 5) are drafted, not decided; the Milan date; and Marc should be on the reply.

In one line. The paper's claim, in plain words. A diffusion model needs, at every noise level, the best guess of the clean data given the noisy data. When the data are a Markov chain (each value depends only on the previous one), that guess can be computed *exactly* from a small model of one step of the chain, and the one-step model can be learned from noisy data alone. Done this way it beats neural denoisers trained on the same sequences, even when the networks see them clean, *provided* the one-step model has the right form: on the physical field of Part II the paper's original form fails and the field's own form wins.

1 The method from zero

The data. Sequences $a_1, \ldots, a_L$ that form a *Markov chain*: given a_i, the next value a_{i+1} does not depend on anything earlier. The law is then fixed by the *transition kernel* $K(a' \mid a)$, the density of the next value given the current one (plus the law of the first site, which for a stationary chain follows from K). The paper's test chain is *linear*: $a_{i+1} = \rho\, a_i + \varepsilon_i$ with non-Gaussian (Laplace) driving noise ε_i.

What a diffusion model needs. It corrupts the data at a level t, $x = e^{-t} a + \sqrt{\Delta_t}\, z$ with $\Delta_t = 1 - e^{-2t}$ and z standard Gaussian noise, and learns the *score* $\nabla_x \log p_t(x)$, where p_t is the density of the corrupted x. The score is used to denoise and, run backwards from pure noise (the "reverse process"), to generate. By Tweedie's

identity

$$\nabla_x \log p_t(x) \;=\; -\frac{x - e^{-t}\,\mathbb{E}[a \mid x]}{\Delta_t},$$

so the score is equivalent to the best denoiser in mean squared error, the posterior mean $\mathbb{E}[a \mid x]$.

Why a chain is special. On a chain the posterior mean can be computed exactly by *belief propagation* (BP), the forward–backward algorithm: one sweep left to right and one right to left, each carrying a "message" (a function over one site's value) through K. We store each message on a grid of 401 points on $[-8, 8]$ ("grid BP"). With the true kernel this gives the reference score, accurate to the grid's quadrature error; we call it the **oracle**.

When the kernel is unknown we learn it from the noisy sequences by *EM* (expectation–maximisation): with the current K, run BP on each noisy sequence to get the posterior of every neighbouring pair (E-step); refit K to those soft pair counts (M-step); repeat. "EM–BP" is this, followed by BP with the learned kernel.

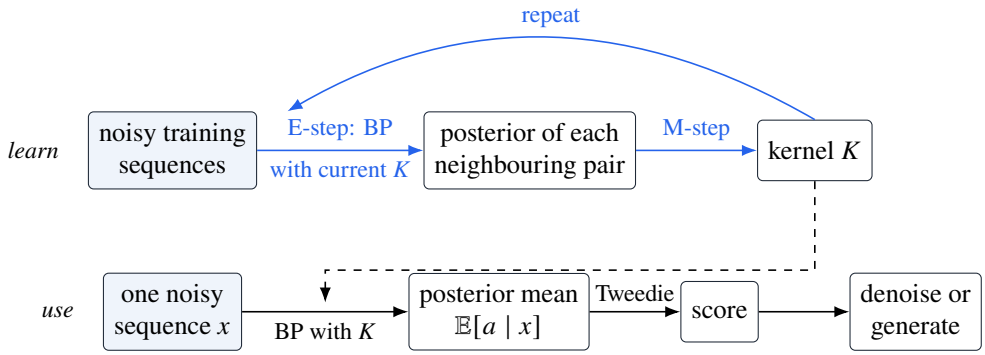


Figure 1: The method. Top: EM learns the kernel from noisy data, with BP inside. Bottom: BP with the learned kernel gives the posterior mean and hence the score. A neural denoiser learns the map from x to the posterior mean (or score) directly instead.

The kernels we fit inside EM–BP (one table for the whole document):

Table 1: Kernel families. The middle three are one model class (a Gaussian step re-weighted by a function of where it lands).

name	what it is	numbers
mixture (the paper's kernel)	next value = $\rho\times$ current + noise drawn from an 8-component Gaussian mixture: the expected next value is *linear* in the current one	25
tilted spline	a Gaussian step re-weighted by a smooth function $w(a')$ of the landing point ($\log w$ a spline)	25
Legendre tilt	the same, $\log w$ a polynomial	7
site energy	the field's own form (Jérôme's pair energy): a Gaussian "bond" between neighbours plus a quartic potential at each site	5
unrestricted	$\log K$ any smooth function of (a, a') (a 10×10 spline); assumes no Gaussian bond	100
MDN	a mixture density network: a small neural net mapping a to a Gaussian mixture for a'	1,516

The competitors. (i) *Neural denoisers* (window CNN, transformer, bidirectional GRU, TCN, U-Net, MLP) trained on the same M sequences. In Part I they see the sequences *clean* and are corrupted afresh at every training step (denoising score matching), so they get strictly more than EM–BP, which sees one noisy copy of each. In Part II they are trained both ways: on one noisy copy ("ISM", EM–BP's own setting) and on clean targets with one fixed corruption ("DSM"). (ii) The **linear baseline** (called the "linear floor" in the reports): the best *linear* denoiser, built from the true covariance. On a linear chain, Gaussian closure of BP (forcing every message to be Gaussian) returns exactly this (the paper's Proposition 1). A method that goes *below* it uses structure beyond second order. In this document "Gaussian" methods always mean this baseline unless said otherwise.

How a method is scored. Its *risk* is a relative RMS error,

$$R = \left(\sum \|\hat{S} - S\|^2 \,/\, \sum \|S\|^2 \right)^{1/2},$$

with $\hat{S}$ its score, S the oracle score, and the sums over test chains, sites and twelve noise levels t from 0.05 to 3. $R = 0$ is perfect and a zero score has $R = 1$. Because it pools sums, the low-noise levels dominate (the six lowest carry 98–99.5% of the squared error for EM–BP and the linear baseline, 88–95% for the networks). M is the number of training sequences; a *seed* is an independent repetition with fresh data and initialisation. For EM–BP and the networks alike, the stopping point is chosen on a validation set by this oracle risk. An oracle-free rule for EM (held-out likelihood) is also recorded: on the paper's chain it picks the same stopping point (7 of 8 settings); on the field it gives somewhat larger risks (Section 3.3).

How the work is kept honest. Experiments since 25 September (E22, E23, …) have a registry file written before the main run, with predictions and the script that will read them; amendments are dated in the file. Outputs are frozen by checksum (three runs of 26 September were frozen late, on 9 October, after being read). A prediction is read by its own rule: CONFIRMED, FALSIFIED, or UNRESOLVED when the rule cannot decide (for example a model still improving when its budget ran out).

2 Part I: Jérôme's eleven questions (call of 25 September)

The test model is the paper's linear chain with $\rho = 0.85$, Laplace (sharp-peaked) driving noise and $L = 32$ sites, unless a question needed another (stated). The questions fall into five themes.

A. Can BP's messages be summarised by a few numbers? (Q1–Q5). This matters because the 401-point grid is where BP spends its time: a parametric family that BP maps onto itself (a "closed" family) would replace the grid.

- **Q1. Gaussian messages = Gaussian score = Kalman smoother?** Yes on a linear chain: the paper's Proposition 1 proves that BP with Gaussian messages returns the best linear denoiser. As a numerical check, on a Gaussian chain the exact score computed three ways (closed form; an independent Kalman smoother, i.e. the Rauch–Tung–Striebel forward–backward filter; our grid BP) agrees to a relative error of 1.9×10^{-13}. *A registered clause failed: BP with messages forced to be Gaussian, run on the same 401-point grid, was off by up to 0.26 in relative score error, because its backward messages widen beyond the grid's edge.* [E39]
- **Q2. With Laplace noise, is BP exact with a Laplace-like message family?** No. We tested Jérôme's family (a Gaussian passed once through the kernel) and a Gaussian+Laplace mixture. Jérôme's family is exact at the first site only (error at machine precision); deeper in the chain its error, the KL divergence to the true message, is about 1.6×10^{-5} at site 31 at low noise, small but not zero, so BP with it is not exact. [E22]
- **Q3. What do the messages look like?** Close to Jérôme's family: at low noise its KL to the true message is 2,272× smaller than a plain Gaussian's and 78× smaller than the 4-parameter Gaussian+Laplace mixture's. *We predicted it would do best at high noise; it does worst there, and the mixture overtakes it.* [E22]
- **Q4. Can any finite family close the recursion?** The Gaussian does (on a Gaussian chain). Jérôme's family does not (in 432 of 432 configurations). We have no proof that no other finite family closes; approximately (refit after each step) any family works, at a measured price. [E22]
- **Q5. BP in Fourier space with ~20 modes (Fourier coefficients)?** As a *representation*, partly: on the Laplace chain a Fourier basis on a window adapted to each message needs 33 numbers (KL 10^{-2}) or 47 (10^{-4}) against the grid's 45 and 151, but it reaches 10^{-4} in only 53 of 144 cases and adding modes can make it worse, so a solver could not choose a truncation and know its error. Merely speeding up the transition step on the grid (by FFT) is capped at about 1.15×, because that step is 12–15% of a BP sweep; an FFT version we built is second-order accurate (6.7×10^{-4} from dense BP) but 5.8× slower at 401 points. The cheap lever turned out to be a coarser grid (Section 3.6). [E23, E24x, E40]

B. Can the kernel be a neural network? (Q6). It can be fitted by EM, and it loses: the MDN is worse than the 25-number mixture kernel in 64 of 64 comparisons (median factor 2.0–2.5), and also worse than the best

plain score network (which has 19–25% lower risk). *We predicted it would sit between the networks and EM–BP.* With 61 times more parameters it does 2–2.5 times worse on test data, consistent with the overfitting Jérôme expected. This verdict is for the paper's chain, where a small kernel is correctly specified; where it is not, the network kernel does win (Section 3.4). [E25m, E30]

C. Cost (Q7, Q11). BP is slower per sequence; by how much depends on threads and grid size. On the paper's 401-point grid with 8 threads BP takes 0.52 ms per sequence against 0.042 ms for the transformer (12.5×); on an H200 GPU 15× at batch 512 and 6× at batch 2048. With one thread the gap narrows, because the networks use threads and BP barely does: at 401 points BP is 3.2 times slower than the transformer and 10.6 times slower than the GRU; at 151 points (0.173 ms) it is faster than the transformer and TCN but still 2.7 times slower than the GRU and 7 times slower than the window network. (The networks timed are the largest screened variants, several times the size of those scored.) Cost grows linearly with the number of sequences and with L; the grid enters like $N_g^{1.2}$, not N_g^2, because elementwise work dominates. [E24x, E47]

D. Sampling without diffusion (Q8). Yes: drawing site 1 and then each next site from the true kernel (ordinary Markov-chain simulation) samples exactly. We compare samplers by the KL divergence between their residual histograms and a large reference sample: exact simulation sits at the Monte Carlo floor, the reverse process with the exact score needs 200 steps to get there, and exact simulation is 731× cheaper (8 threads, 2,048 sequences). With a fitted kernel the samples inherit the kernel's error, the same as the reverse process gives. A Gaussian score generates Gaussian samples: the excess kurtosis of the residual $a_{i+1} - 0.85a_i$ is about 0 against 1.92 for exact samples at the same final noise level (descriptive, no prediction). [E27, E26]

E. How we evaluate (Q9, Q10). Q9: per noise level, EM–BP beats every network at all 12 levels. It beats the linear baseline (which knows the true covariance) at the 8 (with $M = 128$) or 10 ($M = 2048$) lowest levels; above that the remaining non-Gaussian structure is smaller than the fitted kernel's estimation error, and the crossover moves up with data. *We predicted EM–BP would be first at every level.* [E28] Q10: a panel of the paper plots, against t, how much information the noisy data carry about the chain's correlation (a second-order property) and about the shape of its noise (fourth order). Jérôme asked whether the shape curve measures the fourth cumulant κ_4 or the parameter β of the noise law. Recomputed for κ_4, the ratio of fourth- to second-order information falls from 0.071 at $t = 0.05$ to 9.9×10^{-5} at $t = 3$, a factor of about 700, so the paper's message survives (the figure script recomputes it in memory from the frozen E01 file; the derived table itself is not frozen; an earlier figure of 164× used $t = 4$, where the estimator sits at its numerical floor). The old panel plotted β under the label "fourth order", which is off from the κ_4 curve by 1.6–1526×; the rewritten draft re-plots it in cumulant coordinates (Section 4).

Other registered Part I predictions that failed, all reported in the registry: the grid exponent of 2 (it is 1.2), a representation beating the grid by at most 4× *(Fourier on adapted windows does better, unreliably), one exact sweep beating the reverse process on kurtosis (they tie at the Monte Carlo floor), and the FFT breaking even at 3,201 points.*

In one line. On the paper's test model, EM–BP with a 25-number kernel, learned from one noisy copy of each sequence, beats every network (trained on the clean sequences) at every noise level; exact sampling needs no diffusion; per sequence BP is slower than the networks, by 12× on many threads and by 1–3× against the larger networks on one thread with a coarser grid.

3 Part II: the ϕ^4 field

3.1 The model

Think of a one-dimensional elastic string whose height $\phi(s)$ at position s prefers the values +1 or −1, shaken by thermal noise at temperature Θ. Its energy is $H[\phi] = \int \left(\frac{1}{2}\phi'(s)^2 + V(\phi(s))\right) ds$ with the double well $V(\phi) = (\phi^2 - 1)^2/4$, and its dynamics is the stochastic Allen–Cahn equation $\partial_\tau \phi = \partial_s^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta$ (τ is physical time, η space–time white noise). It settles into the Gibbs law $\propto e^{-H[\phi]/\Theta}$. A snapshot (Figure 2) shows *domains* near +1 and −1 separated by *kinks*; the energy of one kink is $E_{\text{kink}} = 2\sqrt{2}/3 \approx 0.94$. When cold, domains are long and the values have a two-humped histogram.

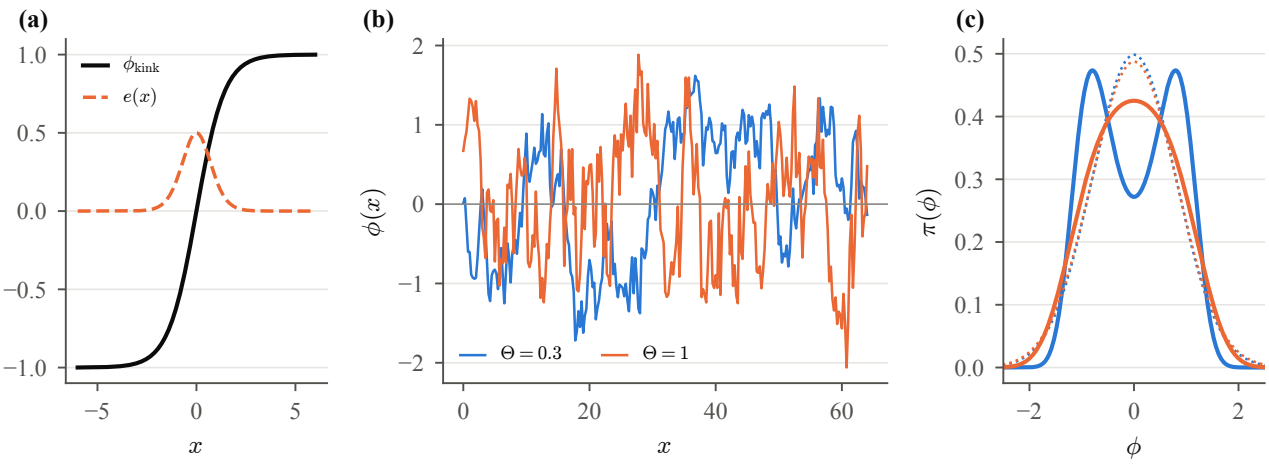


Figure 2: (a) One kink, the field crossing from −1 to +1, and its energy density. (b) Two exact snapshots on a lattice of spacing $h = 0.25$, at $\Theta = 0.3$ (long domains) and $\Theta = 1$. (c) The exact density of the field values (solid) against a Gaussian of the same variance (dotted), $h = 0.25$: two humps when cold.

Why it fits the method exactly. On a lattice of spacing h the energy only couples neighbouring sites:

$$H_h = h \sum_i \left[\tfrac{1}{2}\big((\phi_{i+1} - \phi_i)/h\big)^2 + V(\phi_i)\right],$$

so the Gibbs law factorises along the lattice, and a snapshot $\phi_1, \ldots, \phi_L$ is *exactly* a Markov chain in the site index, and its kernel is a Gaussian step of variance Θh re-weighted by a function of where it lands (a "tilted walk"; Table 1). The site energy contains the truth exactly; the spline and Legendre tilts are flexible members of the same class. Unlike the paper's linear chain, the expected next value is a non-linear function of the current one (on cold cells it pushes sites off the barrier at 0 towards the domain values), so the mixture kernel, which is linear in this sense, cannot represent it. The *correlation length* ξ (over how many sites correlations decay by a factor e) grows fast as Θ falls.

Six test settings ("cells").

Table 2: The cells, and the risk of two reference methods. The iso cells have h chosen so that the slowest correlation mode decays at the same rate as the base cell's (correlation length about 6 sites). The last row ignores both the neighbours and the two humps (it shrinks each value towards 0).

	base	iso	iso	iso	finer	finest
temperature Θ	0.3	0.5	1	2	0.3	0.3
spacing h	1	0.42	0.23	0.15	0.5	0.25
sites L	32	32	32	32	64	128
linear baseline	0.269	0.115	0.058	0.041	0.162	0.098
each site alone with a Gaussian prior	0.674	0.657	0.657	0.656	0.714	0.750

The linear baseline measures how much there is to gain beyond a Gaussian model at each cell. It falls with temperature, and also under refinement, where the histogram stays nearly as two-humped (excess kurtosis −1.35 to −1.22) but neighbouring sites become more redundant. [E32]

3.2 Checking the instruments

- **The oracle is exact**: on every cell grid BP at 401 points agrees with a four-times finer grid to 1.3×10^{-12} (checked before any learning). Later, on a Gaussian chain, grid BP matched two independent exact answers to 1.9×10^{-13}. [E32, E39]
- **Forcing BP's messages to be Gaussian breaks down on the field**: the backward messages become almost flat and their mass piles up at the grid edge. So the Gaussian comparison is the linear baseline itself, computed in closed form. [E32]

- **The paper's information-decay theorem** (Theorem 5 in the rewritten draft: the squared gap to the Gaussian score is a known constant times e^{-8t} at large t) is not reached within the schedule: the measured/predicted ratio at $t = 3$ is 0.85, 0.86, 0.73, 0.94 on the four $L = 32$ cells, and on the $\Theta = 1$ cell it falls over the last step. *Registered as "approaches 1": unresolved.* [E32]
- **Cost** (measured on the paper's chain; BP's cost does not depend on the model): one BP call on 512 sequences of 32 sites needs about 260 MB, five message-sized arrays, linear in everything, 1.8–4.2 times the largest network. [E38]

3.3 Learning the kernel through noise: the main result

Each training sequence enters as one noisy copy at one of the twelve noise levels. EM–BP and the ISM networks never see clean data; the DSM networks also get the clean targets, which favours them, and still lose.

- **The 5-number site energy is the best method on every cell at every size** (on average over seeds). At $M = 2048$ its risk is 3–17% of the linear baseline (4–23% with the oracle-free stopping rule) and 43–59% below the spline; it reads the temperature off the noisy data within 10% in 96 of 96 fits. With very little data on the two hottest cells it still loses to the baseline ($M = 32$: 1.4×); from $M = 128$ it wins on every cell on average (16 of 16 seeds from $M = 512$). *We had predicted it would be within 10% of the spline; it is far better.*
- **The paper's own mixture kernel learns nothing beyond the linear baseline** (1.00–1.02 times it at $M = 2048$): a kernel whose expected next value is linear cannot hold sites in a domain. On non-linear physics the kernel family matters.
- **Networks lose to the spline and the site energy on every cell**: 4–27 times the spline's risk, and above the linear baseline on the hot cells (on the cold cells they do beat the paper's mixture kernel). The window network chose its radius per cell from $\{2, 4, 8\}$ sites and picked 2, 4, 8 at $h = 1, \frac{1}{2}, \frac{1}{4}$, a constant physical reach (at $h = \frac{1}{4}$ the largest offered). Its absolute risk fell (0.149 to 0.118) but relative to the baseline it rose from 0.55 to 1.20 (ISM version): Jérôme's expectation that structure gains under refinement holds, even though the window was free to widen with the lattice. *We had predicted its absolute risk would grow.* Whether an even wider window closes the gap was not tested.
- **The good kernels are physically right**: samples drawn from the spline and the site energy reproduce the kink density and mean domain length within a few per cent, and the whole distribution of domain lengths within 10% at every quantile, statistically indistinguishable from exact samples (4 seeds; one quantile is 10.4% off). On the three cold cells the mixture's samples have about 1.4 times too many kinks and domains 12–18% too short at the median; on the hotter cells they are within about 10%.
- **Two doubts removed.** The MDN's gap to the spline (1.9–4.3×) is not a training-budget artefact: four times the EM budget lowers its risk by only 0.4–4.5% (*we predicted more than 10%*). The potential's barrier height is learnable at high temperature, just slowly: at $\Theta = 2$ its error falls like $M^{-0.54}$ and all 16 seeds are within 20% at $M = 32{,}768$.

Other registered predictions that failed here. Two on substance: with little data at high temperature ($M = 128$, $\Theta = 2$) the paper's mixture beats the spline in 12 of 16 seeds, and the 7-number Legendre tilt crashes numerically on the finest lattice (39 fits); the raw count of sign changes is not a lattice-independent kink density (a coarse count replaced it). The rest on thresholds set too tight: the barrier at $\Theta = 1$ (14 of 16 seeds within 20%, not 16), the domain-length KS test, the tail deficit of the mixture's domains. [E33, E34, E37, E41, E42]

In one line. With the right kernel family, exact BP using a kernel learned from noisy data beats the linear baseline and every network by a wide margin on real non-linear physics; with the paper's linear-drift mixture it collapses to the Gaussian answer.

3.4 When the assumptions break

A hidden variable shared by the whole sequence (on a Gaussian test chain with a closed-form answer, not the field). Each sequence is $b\,G + U$, with U a Gaussian linear chain ($\rho = 0.85$) and $G = \pm 1$ a hidden sign

per sequence; the data are no longer Markov. The paper's first-order kernel (it conditions on one previous site) gets stuck: at $b = 2$ four times more data change its risk by +0.2%, and at $M = 2048$ it is 1.57 ($b = 2$) and 1.67 ($b = 1$) times *worse* than the linear baseline, in 16 of 16 seeds. Running BP on pairs (value, sign), so that it also infers each sequence's sign, repairs it: the risk keeps falling and ends 6.3 times below the baseline at $b = 2$. *We predicted the repair would change nothing when $b = 0$; it helped.* We traced that to the reflection symmetry $x \to -x$: fitting the plain kernel on each sequence and its sign-flipped copy is 32–39% better than the plain fit and 16–17% better than the repaired model (*we predicted equal*), and equal in average risk to fitting on twice as many sequences (*the registered non-inferiority bound failed on interval width: seeds vary a lot*). [E35, E35x, E46]

Observing only every s-th site ("stride"; kernels fitted on clean, noise-free chains). A thinned Markov chain is still Markov (its kernel is K applied s times) but no longer a tilted walk. The coarse-grained physics (kink density, domain length) agrees across strides within 8%. From stride 1 to stride 4 the site energy's and the spline's risks grow about 25 and 14 times (16 seeds), yet stay far below the linear baseline (0.040 and 0.036 against 0.235). Here the flexible network kernel earns its parameters: with a long training budget (16,000 updates) the MDN is below the site energy at stride 4 in 16 of 16 seeds, at about 0.4 times its risk, while at stride 1 the site energy beats it by about 3 times. *We predicted the MDN would settle before its last checkpoint; in 11 of 16 fits it was still improving, so its risks are upper bounds, which only strengthens the stride-4 result.* The noisy-data version under strides was not run. [E36, E36x, E48]

A chain outside the model class. Replace the Gaussian bond by a sharp Laplace bond ($|a' - a|$ instead of $(a' - a)^2$). The site energy and spline nearly stall at a risk of about 0.10, while the unrestricted kernel keeps improving and ends 1.5 times better at $M = 2048$ (8 of 8 seeds). On two true ϕ^4 cells the unrestricted kernel costs 2.5–5.5 times the site energy, and *that price grows with data, the opposite of our prediction*. *At $M = 128$ we predicted the unrestricted kernel would be at least twice the site energy's risk in 14 of 16 seeds on both cells; on the hotter cell it was 12.* So keep the site energy for physics of this gradient type, and use the unrestricted kernel as a misspecification test: if it beats both the site energy and the spline at large M, the Gaussian-bond assumption is wrong. On the same chain the network kernel (the MDN, learned by EM through noise) does better still: at $M = 2048$ its risk is 0.056, against 0.068 for the unrestricted kernel and 0.103 for the site energy (7 of 8 seeds below it, ratio 0.55). *We predicted the 100-number kernel would again beat the network; it does so in 1 of 8 seeds. We also predicted the network would settle within its budget; in 5 of 8 fits it was at its last checkpoint, so its risks are upper bounds.* Together with the stride-4 result, this answers Jérôme's question whether a neural kernel ever earns its parameters: yes, when the simple kernels' class is wrong (by a factor 1.8 to 2.5 against the best of them), not when it is right. One constructed chain and eight seeds. [E43, E49]

Data out of equilibrium. Snapshots taken while the field is still coarsening (domains growing) after a sudden cooling are not Gibbs. Off equilibrium there is no oracle, so the reference is the clean snapshot and the ratios are compressed (a perfect denoiser does not reach zero). Against an empirical Gaussian smoother the site energy's ratio is 0.986 shortly after cooling and 0.950 at equilibrium, below 1 in 8 of 8 seeds at every time. *We predicted the flexible spline would take over off equilibrium; it equals the site energy to three digits.* [E44]

3.5 A new use: missing data with uncertainty

Hide part of a noisy sequence (a random half, or a gap of 8 or 16 of the 32 sites) and draw whole posterior sequences with forward-filtering backward-sampling (FFBS), which gives intervals, not only a mean. The learned site energy and spline match the oracle within 0.4% in error and interval width, with the same coverage (0.89 for a nominal 90% interval). The learned mixture behaves like a Gaussian smoother. A Gaussian smoother with the true covariance is only 1.5–4.6% worse on the cold cell, with 7–9% wider intervals, and at $\Theta = 1$ within 1% in error and 2% in width. *We predicted a visible miscalibration and intervals at least 10% wider; it narrowly over-covers instead, with intervals 9.4% wider.* [E45]

3.6 Making BP cheaper: the grid

Jérôme called the grid the bottleneck. The 401-point grid had only been compared with finer grids; E47 tried coarser ones.

- With the true ϕ^4 kernel, 101 points are already exact to 2×10^{-11} on every cell (smooth integrands).
- The learned ϕ^4 kernels (site energy and spline, refitted on the coarse grid) lose nothing at 101 points (risk equal to the 401-point fits to three digits, on two cells and two data sizes) and at most 1.2% at 51, and their EM fits run about 14 times faster at 101 points.
- The paper's fitted mixture kernels need more, because with lots of data they learn components narrower than a 101-point spacing. At 151 points each kernel family's average risk is within 0.08% of its 401-point value (worst single kernel 0.23%). At 101 points the 8-component mixtures fitted on $M = 2048$ sequences move by 2.2% (worst seed 7.6%; 0.12% at $M = 32$). *We predicted at least 90% of them would be resolved at 101 points; only 22% were (none at* $M = 2048$*).*
- Time per sequence on one thread: 0.053 ms at 51 points, 0.107 at 101, 0.173 at 151, 0.249 at 201, 0.667 at 401.
- The old "BP is 12–18 times slower than networks" figures used 8 threads, which the networks exploit and BP barely does (0.52 ms on 8 threads against 0.67 on one, at 401 points). At one thread and 151 points BP is faster than the largest transformer and TCN, though still slower than the GRU and the window network.

[E47]

In one line. The practical answer to the bottleneck is a coarser grid (4–6 times cheaper, at no loss for the ϕ^4 kernels), not a new Fourier solver.

3.7 Speciation: when does the denoiser commit to a domain?

Jérôme suggested using the temperature sweep to watch "speciation": as noise is removed, the denoiser must at some point *commit* to the sign of each domain. For each site the exact posterior gives $p = P(\text{clean value} > 0)$; the commitment $|2p - 1|$ runs from 0 (undecided) to 1 (certain). E50 records the noise level t_c at which the site-averaged commitment falls through $\frac{1}{2}$ (in the reverse process, "earlier" means at higher noise), and compares with a Gaussian "twin" that has the same correlations but no humps. All five registered predictions on commitment held:

- the posterior's confidence is calibrated: the share of signs guessed right equals $(1 + \text{commitment})/2$ to within 0.003 on average over noise levels (0.016 at worst at one level);
- colder fields commit earlier: t_c falls from 1.61 at $\Theta = 0.2$ to 0.46 at $\Theta = 2$;
- the exact posterior commits earlier than its Gaussian twin at every temperature, by 0.47 in t when coldest and 0.07 when hottest. At $\Theta = 0.2$ it is half-committed when the noise variance is 24 times the signal's; the twin only at 9 times. So the two humps, not only the correlations, decide when the denoiser commits;
- at $\Theta = 0.2$ ($\xi \approx 19$ sites) t_c of the central sites grows with the chain length (1.03 at $L = 4$ to 1.72 at $L = 64$) and stops growing once the chain is several domains long (1.750 and 1.754 at $L = 128$ and 256);
- for chains shorter than ξ, t_c moves one-for-one with $\frac{1}{2}\ln(1 + \Lambda)$, Λ the largest eigenvalue of the covariance (slope 0.98): the finite-chain analogue of the speciation time $\frac{1}{2}\ln\Lambda$ of Biroli, Bonnaire, de Bortoli and Mézard (2024).

A second part checked the lattice against the continuum. As $h \to 0$ the physical correlation length converges (it moves by at most 0.04% from $h = 1/16$ to $1/32$; the $h = 1$ lattice overestimates it by 4–9.5%). *We predicted that a fit of* $\ln\xi = E_{\text{kink}}/\Theta + \frac{1}{2}\ln\Theta + \ldots$ *on* $\Theta \in [0.1, 0.3]$ *would return the kink energy* $2\sqrt{2}/3 \approx 0.943$ *within 2%; it returns 0.988, and 0.93 instead of* $\frac{1}{2}$. The local slope approaches the theory as Θ falls, with a gap shrinking roughly like Θ^2, so the law looks right but these temperatures are not cold enough to pin its constants. [E50]

4 Where things stand

Nothing is running. The last three cluster experiments ran on 10 October, after the cluster's maintenance window, and are frozen and read: the learned kernels on a coarse grid (E47b), the network kernel under strides

(E48) and the network kernel on the out-of-class chain (E49), all in Section 3. The laptop run E50 finished the same day.

Answered by frozen evidence. Jérôme's eleven questions (Q5, Q8 and Q10 with the caveats above), and the note's programme: the four-arm comparison on the field, its physical checks (correlations, kinks, domain lengths, bimodality, generated samples), runtime and memory, the Gaussian/Kalman sanity check, the FFT transition, posterior sampling, clean-data learning, the hidden-variable repair, the subsampling protocol (completed by E48), the coarse grid (E47), the neural kernel's range of use (E48, E49) and speciation (E50).

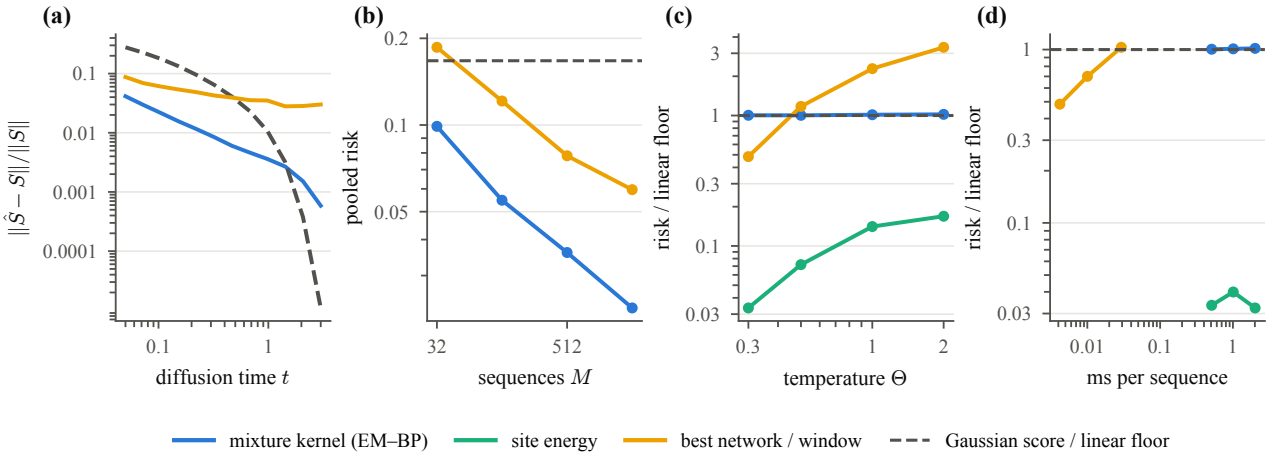


Figure 3: The headline in one picture. (a, b) Part I, the paper's chain: error per noise level at $M = 2048$, and error against M. The learned kernel (blue) beats the best network (yellow; trained on clean sequences) at every level and size, and the Gaussian score (dashed) except at the highest noise levels in (a), $t \gtrsim 2$. (c) Part II, the ϕ^4 field: risk divided by the linear baseline (dashed) across temperature at $M = 2048$. The site energy (green) is far below it, the paper's mixture kernel (blue) sits on it, the window network (yellow; DSM, i.e. trained with clean targets) rises above it when hot. (d) The same ratio against inference time (ms per sequence, 8 threads, largest network variants) on three lattice spacings at $\Theta = 0.3$.

The paper (12 October). The draft was rewritten to follow Jérôme's and Marc's note: it opens with the structural message (learn the local law, infer globally) and a summary figure; states everything for a general Markov kernel, with the linear chain as the first example and the ϕ^4 field as the second; has a Gaussian sanity-check section (BP with Gaussian messages is the linear score and the Kalman smoother); presents the Fisher/Hermite theory as a supporting subsection rather than the headline; gives the field its own section with the temperature sweep, the scoreboard, the physical reading of the fitted kernel, runtime and speciation; and has a cost section with the coarse grid and the reasons not to use Fourier modes. Every number in it is generated from the frozen outputs, and the consistency checks and the independent audit pass. It is 39 pages against an 8-page body limit, on purpose: the cut is yours, and `docs/PAPER_REWRITE_2026-10-12.md` lists what to keep (summary, method, Gaussian sanity, per-level gain, sample efficiency, field sweep, accuracy against cost) and what can go to the appendix.

An internal audit (9 October) re-derived the numbers in the registry, the reports and the question ledger from the frozen files. It changed no existing verdict, but it found three runs never frozen (now frozen), gave first verdicts to predictions never read (one falsified, one unresolved), replaced numbers that had been quoted from unfrozen laptop runs, and flagged speciation as not addressed (hence E50). The corrections are applied, each logged in its registry file; this document uses the corrected numbers.

Drafted in the paper, waiting for your approval.

1. The framing Jérôme suggested on 30 September (a general Markov kernel; the linear chain first, the field second) and the field in the body rather than the appendix, which is what makes the paper long.
2. The information panel (Q10) is re-plotted in cumulant coordinates (κ_2, κ_4); the alternative was to keep β and drop the label "fourth order".
3. The figures: the summary figure plus about six key ones; the candidates, with a suggestion for each, are in Section 5.

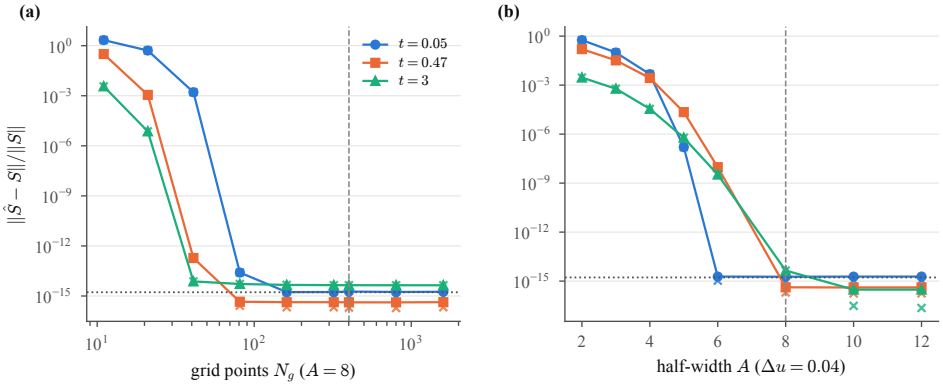
Yours to do. The reply to Jérôme *and Marc*, this week (12–16 October); tell Jérôme the Milan date once it is confirmed.

5 The figures to choose from

Jérôme and Marc's note asks for five or six decisive figures and a summary figure *before* the prose, and Jérôme asked for three specific plots: the information panel with the right labels (Q10), the speciation experiment, and the cost. The summary figure (Figure 3, Section 4) and the method figure (Figure 1) are already shown above. Below are the other candidates, each with the question it answers, what to look at, and my suggestion. Under the 8-page limit you can keep the summary plus about six; the rest go to the appendix. The suggestions are mine; the choice is yours.

A. The Gaussian sanity check

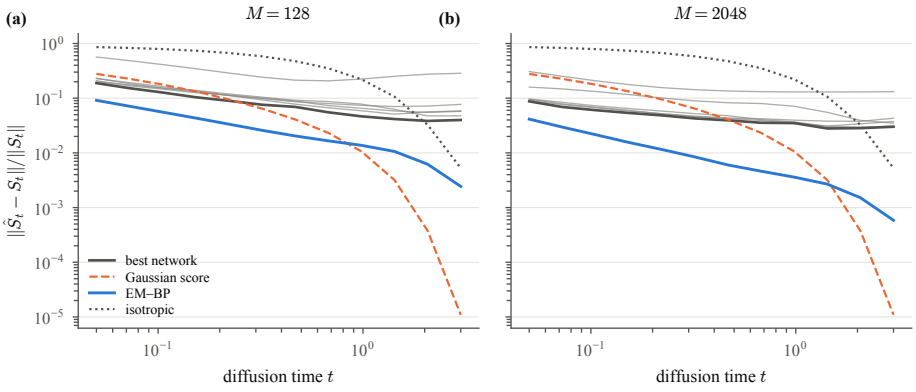
Jérôme's second contribution; Q1; E39



Grid BP against the closed-form score (lines) and an independent Kalman smoother (crosses) on a Gaussian chain, at three noise levels. Look for the error falling as the grid is refined (a) and the domain widened (b), and stopping at 10^{-13}–10^{-15}, the precision floor of the computer. It shows the instrument is right before anything is learned. **Suggest: body (one panel is enough).**

B. Where the gain is, level by level

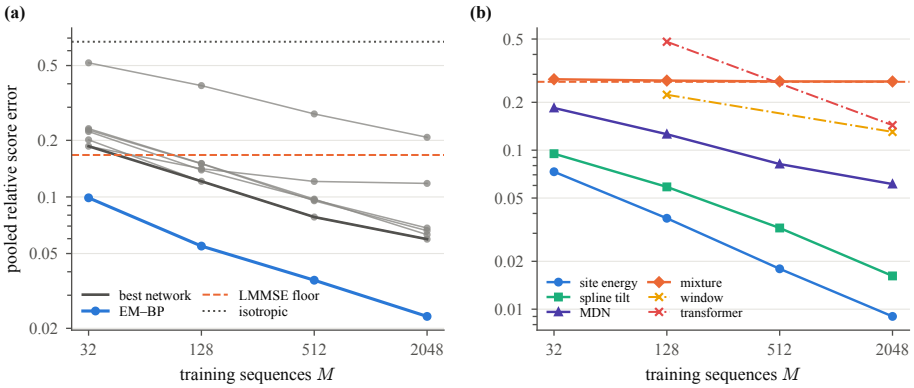
Q9; E28



Error against the exact score at each noise level on the Laplace chain, with 128 (a) and 2048 (b) training sequences. The learned kernel (blue) is below every network at every level, and below the Gaussian score until the noise has erased the non-Gaussian structure (the crossing moves from $t \approx 1$ to $t \approx 2$ with more data). **Suggest: body.**

C. Sample efficiency

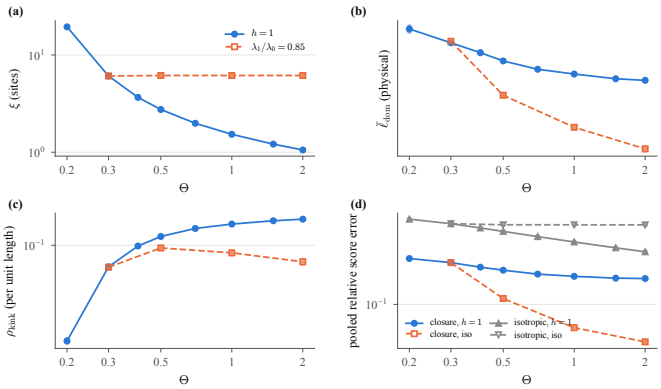
the note's figure 4; E02a, E03, E33



Error against the number of training sequences. (a) On the Laplace chain, EM–BP with 128 sequences (0.055) is already better than the best network with 2048 (0.060). (b) On the ϕ^4 field the site energy improves steadily while the paper's mixture kernel sits on the dashed linear floor. **Suggest: body.**

D. The field and the temperature sweep

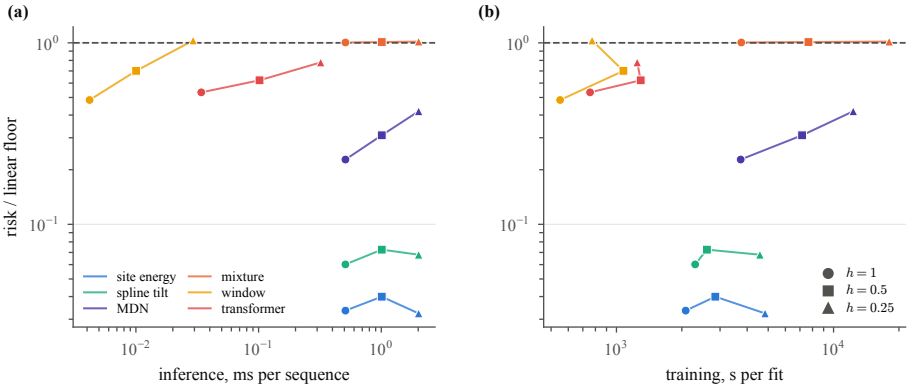
the note's field programme; E32, E33



How the ϕ^4 field changes with temperature Θ for the fixed spacing $h = 1$ (blue) and for spacings chosen to keep the correlation length fixed (orange): correlation length, domain length, kink density, and (d) how much there is to gain beyond a Gaussian model (the linear closure) and an independent-sites model. Cold fields are the nonlinear, domain-forming ones. **Suggest: body (it introduces the second laboratory).**

E. Accuracy against cost

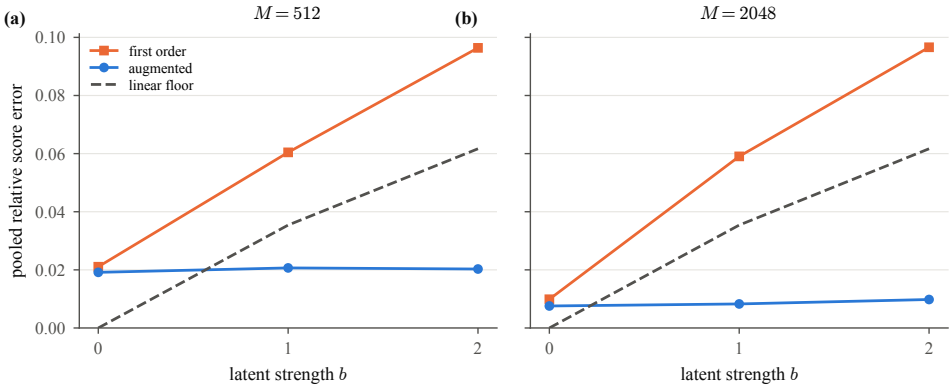
Jérôme's cost question (Q7); E24x, E33



Error divided by the linear floor against inference time per sequence (a) and training time (b), on three lattice spacings. The networks own the cheap end and the structured kernels the accurate end; nothing lies between. This is the honest picture of what exactness costs. **Suggest: body.**

F. A hidden variable and its repair

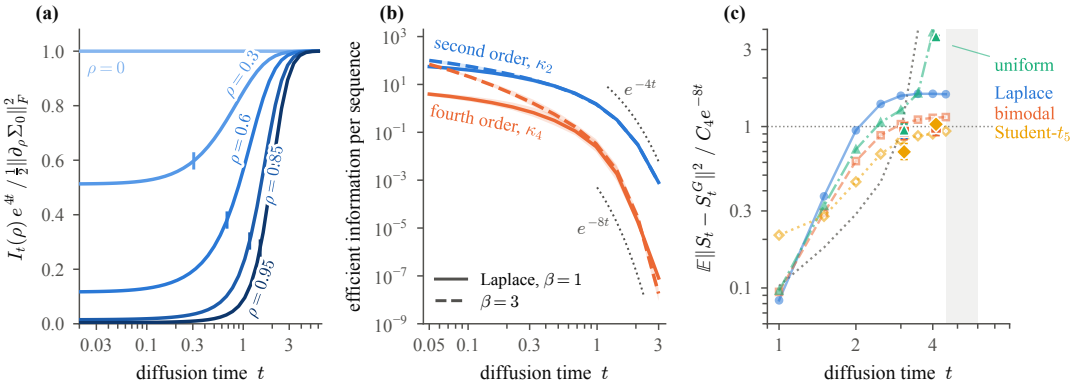
where the method stops; E35x



A hidden sign shared by the whole sequence makes the data non-Markov. The first-order kernel (orange) stops improving with data and ends above the linear floor (dashed); the same kernel on the augmented state (blue) keeps improving. Sixteen seeds. **Suggest: body if there is room, otherwise appendix.**

G. The information panel, in cumulant coordinates

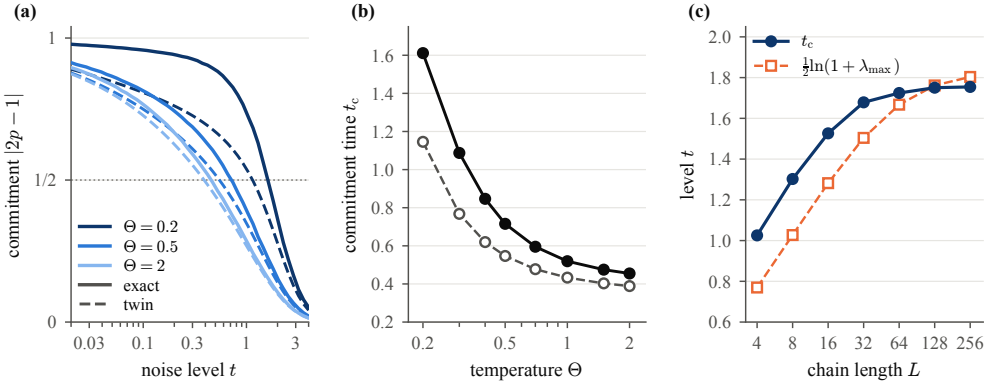
Jérôme's Q10 and action item 4; E01



The panel Jérôme asked to relabel is (b). It now shows the information the noisy data carry about the noise's second and fourth cumulants (κ_2, κ_4), not about ρ and β: fourth-order information (lower curves) decays like e^{-8t} against e^{-4t} for second order; the rate is read on the smooth noise law ($\beta = 3$, dashed) because the Laplace curve reaches the numerical floor. (a) is the Gaussian chain's information about ρ; (c) is the gap between the exact score and the Gaussian one. **Suggest: appendix, unless you keep the information theory in the body.**

H. Speciation: when the denoiser commits

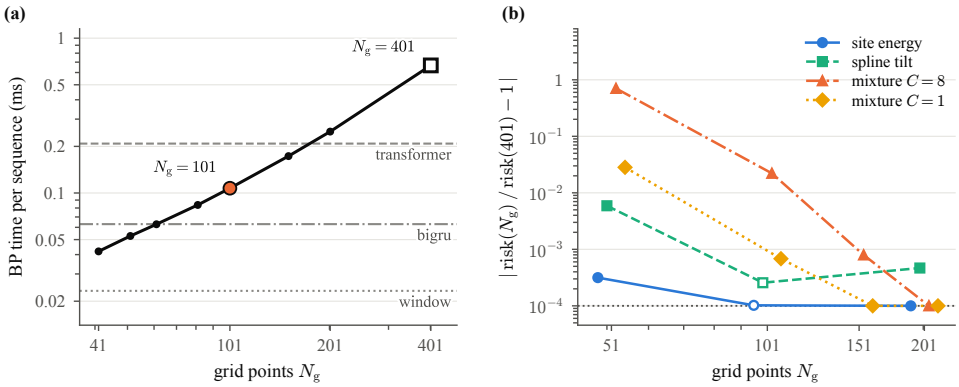
Jérôme's suggestion; E50



(a) The commitment $|2p - 1|$ of the exact posterior falls from one to zero as noise grows; the dashed twin has the same correlations but no humps. (b) The noise level t_c at which it is half-committed falls from 1.61 to 0.46 as the field warms, always later than the twin's. (c) t_c grows with the chain length and then saturates. **Suggest: appendix, with one sentence in the body.**

I. A coarser grid

the grid as the bottleneck (Q7, Q11); E47



(a) BP time per sequence on one thread against the number of grid points, with three networks for scale. (b) How much the fitted kernels' error changes when the grid is coarsened: nothing visible down to 101 points for the field kernels; the linear chain's mixtures need about 151. **Suggest: appendix, or fold (a) into figure E.**

My suggested six for the body, after the summary figure: A, B, C, D, E and F, plus the small method schematic (Figure 1 of this document). G, H and I go to the appendix, with one sentence each in the text. `docs/PAPER_REWRITE_2026-10-12.md` has the cutting plan; the figures themselves are all already in the draft.

Part III The mathematics

This part contains the mathematics behind everything above, in the order in which it is used: the channel and the score (§6), exact inference on a chain (§7), the Gaussian special case (§8), learning by EM (§9), the field (§10), what noise erases (§11), where the structure stops (§12), speciation (§13), cost and numerics (§14), and the experimental protocol (§15). Each result is stated, then proved or sketched. Notation: $\mathbb{E}$ is expectation, $\mathcal{N}(x; m, v)$ the Gaussian density in x with mean m and variance v, $\|\cdot\|_F$ the Frobenius norm, bold letters are vectors of length L.

6 The channel, the score and Tweedie's identity

Setting. A clean sequence $\mathbf{a} = (a_1, \dots, a_L) \in \mathbb{R}^L$ has density

$$P_0(\mathbf{a}) = \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}), \tag{1}$$

with initial law μ and transition kernel K. The *linear chain* is $K(a' \mid a) = p_\varepsilon(a' - \rho a)$ with a driving-noise density p_ε of mean 0 and variance $q = 1 - \rho^2$, and $\mu = \mathcal{N}(0, 1)$; then every a_i has variance 1 and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$, the matrix Σ_0. The Laplace chain takes $p_\varepsilon(e) = \frac{1}{2b} e^{-|e|/b}$ with $2b^2 = q$. At noise level $t \geq 0$ the data are observed through the coordinatewise Ornstein–Uhlenbeck channel

$$\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, I_L). \tag{2}$$

This is the solution at time t of $\mathrm{d}x = -x\,\mathrm{d}s + \sqrt{2}\,\mathrm{d}W$ started at a: multiplying by e^s gives $\mathrm{d}(e^s x) = \sqrt{2}e^s \mathrm{d}W$, so $x(t) = e^{-t}a + \sqrt{2}\int_0^t e^{-(t-r)}\mathrm{d}W(r)$, Gaussian with variance $2\int_0^t e^{-2(t-r)}\mathrm{d}r = \Delta_t$. Its stationary law is $\mathcal{N}(0, 1)$, which is why pure noise is where the channel ends. Write P_t for the law of $\mathbf{x}$ and $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ for its score.

Tweedie's identity (proof). $P_t(\mathbf{x}) = \int P_0(\mathbf{a})\, \mathcal{N}(\mathbf{x}; e^{-t}\mathbf{a}, \Delta_t I)\, \mathrm{d}\mathbf{a}$, and the Gaussian factor $\varphi = \mathcal{N}(\mathbf{x}; e^{-t}\mathbf{a}, \Delta_t I)$ has $\nabla_{\mathbf{x}}\varphi = -\frac{\mathbf{x} - e^{-t}\mathbf{a}}{\Delta_t}\,\varphi$. Dividing ∇P_t by P_t turns the integral into a posterior expectation:

$$S(\mathbf{x}, t) = -\frac{\mathbf{x} - e^{-t}\,\mathbb{E}[\mathbf{a} \mid \mathbf{x}]}{\Delta_t}, \qquad \text{equivalently} \qquad \mathbb{E}[\mathbf{a} \mid \mathbf{x}] = e^{t}\big(\mathbf{x} + \Delta_t S(\mathbf{x}, t)\big). \tag{3}$$

So the score is a rescaled *posterior mean*, the best denoiser in mean square, and a score error is a denoising error multiplied by e^{-t}/Δ_t: an estimate $\hat{S}$ corresponds to the denoiser $\hat{\mathbf{m}} = e^t(\mathbf{x} + \Delta_t \hat{S})$ and $\|\hat{S} - S\| = \frac{e^{-t}}{\Delta_t}\|\hat{\mathbf{m}} - \mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|$. Everything below is therefore about computing, or learning to compute, $\mathbb{E}[a_i \mid \mathbf{x}]$.

7 Exact inference on a chain: belief propagation

The posterior is a chain. By Bayes' rule and the independence of the noise across sites,

$$p(\mathbf{a} \mid \mathbf{x}) \;\propto\; \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}) \prod_{i=1}^{L} \ell_i(a_i), \qquad \ell_i(a) = \mathcal{N}(x_i; e^{-t}a, \Delta_t). \tag{4}$$

Each factor involves one or two neighbouring sites, so the factor graph is a path: a tree. On a tree the sum-product algorithm is exact.

Messages. With $\overrightarrow{m}_1 = \mu$ and $\overleftarrow{m}_L \equiv 1$,

$$\overrightarrow{m}_{i+1}(a') = \int K(a' \mid a)\, \ell_i(a)\, \overrightarrow{m}_i(a)\, \mathrm{d}a, \qquad \overleftarrow{m}_i(a) = \int K(a' \mid a)\, \ell_{i+1}(a')\, \overleftarrow{m}_{i+1}(a')\, \mathrm{d}a'. \tag{5}$$

Proof of correctness. By induction and the Markov property, $\overrightarrow{m}_i(a)\ell_i(a) \propto p(a_i{=}a,\, x_{1:i})$ and $\overleftarrow{m}_i(a) \propto p(x_{i+1:L} \mid a_i{=}a)$: the first claim is Bayes' rule plus the transition integral, the second uses that $x_{i+1:L}$ depends on a_i only through a_{i+1}. Since $x_{1:i}$ and $x_{i+1:L}$ are conditionally independent given a_i,

$$p(a_i \mid \mathbf{x}) \;\propto\; \overrightarrow{m}_i(a_i)\, \ell_i(a_i)\, \overleftarrow{m}_i(a_i), \tag{6}$$

$$p(a_{i-1}, a_i \mid \mathbf{x}) \;\propto\; \overrightarrow{m}_{i-1}(a_{i-1})\, \ell_{i-1}(a_{i-1})\, K(a_i \mid a_{i-1})\, \ell_i(a_i)\, \overleftarrow{m}_i(a_i), \tag{7}$$

and the evidence is $P_t(\mathbf{x}) = \int \overrightarrow{m}_i \ell_i \overleftarrow{m}_i \,\mathrm{d}a$ for any i. The posterior mean $\mathbb{E}[a_i \mid \mathbf{x}] = \int a\, p(a_i{=}a \mid \mathbf{x})\mathrm{d}a$ and Tweedie's identity (3) give the exact score of a correlated, possibly strongly non-Gaussian, L-dimensional law, for any kernel K, even when that score has no closed form. One forward and one backward sweep return all site marginals (6), all pair marginals (7) (which EM needs, §9) and the evidence.

Grid implementation, cost and what "exact" means. Messages are stored on a uniform grid $u_1, \dots, u_{N_g}$ on $[-A, A]$ (here $N_g = 401$, $A = 8$) with composite trapezoid weights w_k, so $\overrightarrow{m}_{i+1}(u_k) = \sum_l w_l K(u_k \mid u_l)\ell_i(u_l)\overrightarrow{m}_i(u_l)$: a matrix–vector product of cost N_g^2 per site. The cost per sequence and level is $O(LN_g^2)$, against $O(N_g^L)$ for brute-force summation. The recursion is run in the linear domain with the discarded scale factors accumulated, so the log-evidence is exact to floating point. Three levels of exactness are kept apart: (i) sum-product is exact on functions; (ii) forward–backward is exact for the *discretised* factor model; (iii) the discretised model approximates the continuous chain with two errors, truncation (mass beyond $\pm A$) and quadrature (second order in the spacing for the trapezoid rule, but faster than any power of the spacing for smooth, rapidly decaying integrands, which is what is measured: errors of 10^{-12}–10^{-15} on the Gaussian chain). Both are measured, not assumed (§14). "The oracle" is BP with the true kernel.

Sampling. Because the kernel is explicit once learned, *ancestral sampling* is exact: $a_1 \sim \mu$, then $a_i \sim K(\cdot \mid a_{i-1})$ (on the grid, an inverse-CDF draw per step). *Forward-filter backward-sample* draws from the posterior given $\mathbf{x}$: a_L from the filtered marginal, then a_i from $p(a_i \mid a_{i+1}, x_{1:i}) \propto K(a_{i+1} \mid a_i)\, \overrightarrow{m}_i(a_i)\ell_i(a_i)$, which needs only the forward messages. A hidden block of sites is handled by dropping the corresponding ℓ_i factors.

8 The Gaussian case: closure, LMMSE and the Kalman smoother

Let Σ_0 be the covariance of $\mathbf{a}$ and $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$. The *covariance twin* is the Gaussian $\mathcal{N}(0, \Sigma_0)$. **Gaussian facts (proof).** $(\mathbf{a}, \mathbf{x})$ is jointly Gaussian with $\mathrm{Cov}(\mathbf{x}) = \Sigma_t$ and $\mathrm{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t}\Sigma_0$, hence

$$\mathbb{E}_{\mathrm{G}}[\mathbf{a} \mid \mathbf{x}] = e^{-t}\Sigma_0\Sigma_t^{-1}\mathbf{x}, \qquad S^{\mathrm{G}}(\mathbf{x}, t) = -\Sigma_t^{-1}\mathbf{x}, \tag{8}$$

and indeed $\mathbf{x} - e^{-t}\mathbb{E}_{\mathrm{G}}[\mathbf{a} \mid \mathbf{x}] = (\Sigma_t - e^{-2t}\Sigma_0)\Sigma_t^{-1}\mathbf{x} = \Delta_t\Sigma_t^{-1}\mathbf{x}$, which is (3). For *any* law with covariance Σ_0, the first expression is also the best *linear* denoiser (LMMSE): the normal equations $\mathbb{E}[(\mathbf{a} - G\mathbf{x})\mathbf{x}^\top] = 0$ give $G = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}$, which depends on the law only through Σ_0. This is the *linear floor* of the reports: the best any second-order estimator can do.

Proposition 1 (Gaussian closure computes the LMMSE). Run the recursion (5) for the linear chain with $\rho \neq 0$, but after every transition integral replace the message by the Gaussian with the same mean and variance. Then the posterior means equal $e^{-t}\Sigma_0\Sigma_t^{-1}\mathbf{x}$, whatever the driving-noise density, so the resulting score is S^{G}.
Sketch. Induction on the sweep. If the incoming message is Gaussian with mean m' and variance v' (after multiplication by ℓ_i, precision-weighted exactly as in the Gaussian chain), then the transition integral is the density of $\rho a + \varepsilon$ with a Gaussian, whose mean and variance are $(\rho m', \rho^2 v' + q)$ for every ε of mean 0 and variance q; projection keeps exactly these two numbers. Backwards, the message is $h(\rho a)$ with h the density of $a'' - \varepsilon$ under the incoming Gaussian, of mean m' and variance $v' + q$, and as a function of a its normalised moments are $(m'/\rho, (v' + q)/\rho^2)$, which is where $\rho \neq 0$ is used. So every projected belief is that of the covariance twin. (The hypothesis "initial message Gaussian or projected" cannot be dropped: a lemma stated without it was false, because the projection does not commute with the likelihood tilt.)

Proposition 2 (BP on the Gaussian chain is the Kalman smoother). Let the driving noise be Gaussian. Then (i) the forward messages are the Kalman *predictions* and $\overrightarrow{m}_i\ell_i$ the filtered densities, (ii) $\overleftarrow{m}_i$ is, up to scale, the likelihood of the future in information form, and (iii) $\overrightarrow{m}_i\ell_i\overleftarrow{m}_i$ is the Rauch–Tung–Striebel (RTS) smoothed marginal; its mean is (8).

Proof of (i). Suppose $\overrightarrow{m}_i = \mathcal{N}(m, P)$. Multiplying by ℓ_i and normalising is Bayes' rule with a scalar observation: the update $m^+ = m + k(x_i - e^{-t}m)$, $P^+ = (1 - ke^{-t})P$ with gain $k = e^{-t}P/(e^{-2t}P + \Delta_t)$. The transition integral then gives the prediction $m' = \rho m^+$, $P' = \rho^2 P^+ + q$ (Proposition 1's moment map, now exact). (ii) and (iii): $\overleftarrow{m}_i(a) \propto p(x_{i+1:L} \mid a_i{=}a)$ is the exponential of a quadratic in a (zero precision at $i = L$), and the product with the filtered density is $p(a_i \mid \mathbf{x})$ by the conditional independence above. The RTS smoother computes the same conditional law by a different factorisation (conditioning backwards on a_{i+1}); two exact computations of one Gaussian law agree. □ **Consequences.** BP is a *smoother* (every site is conditioned on the whole sequence), not a filter. On the Gaussian chain everything is available in closed form, which is why it is the sanity check for any approximate BP: the closed form, an independent RTS smoother and grid BP agree to 1.9×10^{-13} relative score error at every level (10^{-15} for smoother against closed form). The three identifications (closure = LMMSE, LMMSE = Gaussian score, BP = Kalman) are not new (they are in the thesis); what the paper adds is stating them as one proposition. **Where closure fails.** If the messages are forced to be Gaussian *on the grid*, the backward messages (whose variance grows like e^{2t}) are truncated by the domain, and the error reaches 0.26; widening the domain to $A = 64$ removes it up to $t = 1.4$. On the ϕ^4 field the backward messages are nearly flat, with a large share of their mass in the outer 5% of the grid, so the Gaussian comparison there is the closed-form LMMSE itself.

9 Learning the kernel through noise: EM

Objective. We observe M noised sequences $\mathbf{x}^\mu$, each at one level t_μ, never the clean chain, and maximise the marginal log-likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_{t_\mu,\theta}(\mathbf{x}^\mu)$ of a parametrised kernel K_θ, where $P_{t,\theta}(\mathbf{x}) = \int P_\theta(\mathbf{a}) \prod_i \ell_i(a_i)\, \mathrm{d}\mathbf{a}$ is exactly the evidence that BP returns.

EM and its monotonicity (proof). Let $Q(\theta \mid \theta') = \mathbb{E}_{\theta'}[\log P_\theta(\mathbf{a}) \mid \mathbf{x}]$ (summed over sequences). By Jensen's inequality, $\log P_{t,\theta}(\mathbf{x}) - \log P_{t,\theta'}(\mathbf{x}) = \log \mathbb{E}_{\theta'}\big[\frac{P_\theta(\mathbf{a})}{P_{\theta'}(\mathbf{a})} \,\big|\, \mathbf{x}\big] \geq Q(\theta \mid \theta') - Q(\theta' \mid \theta')$, so any θ that increases Q increases $\mathcal{L}$. EM alternates the E-step (compute $Q(\cdot \mid \theta')$ by BP) and the M-step (increase it); increasing without maximising is *generalised* EM and keeps the guarantee.

On a chain, Q depends on the data only through pair marginals. Since $\log P_\theta(\mathbf{a}) = \log \mu_\theta(a_1) + \sum_{i \geq 2} \log K_\theta(a_i \mid a_{i-1})$, the expectation under the posterior uses only (6) at site 1 and the pair marginals (7):

$$Q(\theta \mid \theta') = \sum_{k,l} \Xi_{kl} \log K_\theta(u_k \mid u_l) + (\text{first-site term}), \qquad \Xi_{kl} = \sum_\mu \sum_{i \geq 2} P_{\theta'}(a_i{=}u_k, a_{i-1}{=}u_l \mid \mathbf{x}^\mu), \tag{9}$$

the expected number of transitions from u_l to u_k.

Fisher's identity (proof). Write $P_{t,\theta}(\mathbf{x}) = \int f_\theta(\mathbf{a})\mathrm{d}\mathbf{a}$ with $f_\theta = P_\theta(\mathbf{a}) \prod \ell_i$. Then $\nabla_\theta \log \int f_\theta = \int \nabla_\theta f_\theta / \int f_\theta = \mathbb{E}[\nabla_\theta \log f_\theta \mid \mathbf{x}]$, so

$$\nabla_\theta \mathcal{L}(\theta) = \sum_{k,l} \Xi_{kl}(\theta)\, \nabla_\theta \log K_\theta(u_k \mid u_l) \,\Big(+ \text{first-site term}\Big), \tag{10}$$

with Ξ computed at the *current* θ. The exact gradient of the evidence therefore needs no differentiation through the recursion and no finite differences; it was verified against finite differences and brute-force enumeration on short chains.

The M-steps, family by family.

- *Mixture kernel* $K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c\, \mathcal{N}(a' - \rho a - \nu_c;\, s_c^2)$: a linear drift ρa with Gaussian-mixture driving noise (25 numbers at $C = 8$). Write $S_{00} = \sum_{kl} \Xi_{kl} u_l^2$, $S_{01} = \sum_{kl} \Xi_{kl} u_k u_l$, $S_{11} = \sum_{kl} \Xi_{kl} u_k^2$, $W = \sum_{kl} \Xi_{kl}$. For $C = 1$ the M-step is a weighted least squares: minimising $\sum \Xi_{kl}(u_k - \rho u_l)^2$ gives $\hat{\rho} = S_{01}/S_{00}$ and then $\hat{q} = \sum \Xi (u_k - \hat{\rho} u_l)^2 / W = (S_{11} - \hat{\rho} S_{01})/W$ (using $\hat{\rho} S_{00} = S_{01}$): the belief-weighted Yule–Walker estimator. For $C > 1$ the residuals $u_k - \rho u_l$ carry a second latent label; we alternate conditional maximisation of ρ and of the mixture parameters, a generalised EM.
- *Tilted walks* $K_\theta(a' \mid a) = \mathcal{N}(a'; a, \gamma)\, e^{\theta \cdot B(a')} / Z_\theta(a)$ with B a Legendre (7 numbers) or spline (25) basis. For each source value a this is an exponential family in θ with log-partition $\log Z_\theta(a)$, so $\nabla_\theta \log K_\theta(a' \mid$

$a) = B(a') - \mathbb{E}_\theta[B(a') \mid a]$ and $\nabla_\theta Q = \sum_{kl} \Xi_{kl}\big(B(u_k) - \mathbb{E}_\theta[B \mid u_l]\big)$: the M-step matches the expected sufficient statistics under the posterior to those of the model (Boltzmann learning without a global partition function). The Hessian is $-\sum_{kl} \Xi_{kl} \operatorname{Cov}_\theta(B \mid u_l) \preceq 0$, so Q is concave in θ at fixed γ.

- *Site energy* (5 numbers): the pair energy $W(a, a') = (a' - a)^2/2\gamma + f(a) + f(a')$ with f a polynomial without constant term, of degree four, and $\log \gamma$ as the other parameter. The kernel is built from the symmetric operator $\mathsf{T}_f(a, a') = e^{-f(a)} \mathcal{N}(a'; a, \gamma) e^{-f(a')}$ and its Perron eigenpair (λ_0, ψ_0) by the ground-state transform of §10, so each evaluation needs one eigensolve of an $N_g \times N_g$ matrix, and the gradient needs the derivative of the Perron vector: differentiating $\mathsf{T}\psi_0 = \lambda_0\psi_0$ and projecting on ψ_n, $n \geq 1$, gives $\mathrm{d}\psi_0 = \sum_{n\geq 1} \psi_n \langle \psi_n, \mathrm{d}\mathsf{T}\,\psi_0\rangle/(\lambda_0 - \lambda_n)$ (the component along ψ_0 vanishes because ψ_0 stays normalised; the denominators are positive by Perron–Frobenius). The objective is not concave, so a step is kept only if Q increases.
- *Unrestricted* (100 numbers): $\log K$ a free two-dimensional spline of (a, a'), normalised per column, assuming no Gaussian bond. *MDN* (1,516): a small network mapping a to the weights, means and scales of a Gaussian mixture for a', fitted by the same EM with gradient steps in the M-step.

Every kernel is initialised the same way, run on a ladder of checkpoints (10 to 1200 updates, longer for the network kernels), and the checkpoint is chosen on a validation bundle disjoint from the test bundle, either by the held-out risk against the exact score (needs the truth) or by the held-out one-view log-evidence $\sum_{\mathbf{x}} \log P_{t,\theta}(\mathbf{x})$ (needs no truth; it picks the same checkpoint in 7 of 8 settings on the paper's chain).

The comparison networks. *Denoising score matching* (DSM) regresses $s_\theta(\mathbf{x}_t, t)$ onto $-(\mathbf{x}_t - e^{-t}\mathbf{a})/\Delta_t$, the score of the *conditional* law given the clean sequence, using the clean sequence $\mathbf{a}$; its minimiser is S because $S = \mathbb{E}[-(\mathbf{x} - e^{-t}\mathbf{a})/\Delta_t \mid \mathbf{x}]$ by (3). *Implicit score matching* (ISM, Hyvärinen 2005) minimises $\mathbb{E}[\frac{1}{2}\|s_\theta\|^2 + \nabla\cdot s_\theta]$ over noised samples only; by integration by parts it equals $\frac{1}{2}\mathbb{E}\|s_\theta - S\|^2 - \frac{1}{2}\mathbb{E}\|S\|^2$, so it has the same minimiser without clean data, the same information EM–BP uses.

How a method is scored. The *risk* is the schedule-pooled relative score error

$$R = \Big(\frac{\sum_t \sum_{j,i} \big(\hat{S}_t(\mathbf{x}_j)_i - S_t(\mathbf{x}_j)_i\big)^2}{\sum_t \sum_{j,i} S_t(\mathbf{x}_j)_i^2}\Big)^{1/2}, \tag{11}$$

over test sequences j, sites i and the twelve levels, with S the oracle score; $R = 0$ is exact and the zero score has $R = 1$. The two reference scores are the LMMSE floor $-\Sigma_t^{-1}\mathbf{x}$ (8) and the *isotropic* score $-\mathbf{x}$ (each site alone with its marginal $\mathcal{N}(0, 1)$, for which $\Sigma_t = I$). Because the sums are pooled, the low-noise levels dominate.

10 The ϕ^4 field as a Markov chain

Gibbs law (derivation). For $\mathrm{d}X = -M\nabla\mathcal{H}(X)\,\mathrm{d}s + \sqrt{2\Theta M}\,\mathrm{d}W$ with mobility $M > 0$, the stationary law has zero probability current in every coordinate, which gives $p \propto e^{-\mathcal{H}/\Theta}$ (for one coordinate: $J = -V'p - \Theta p' = 0 \Rightarrow p \propto e^{-V/\Theta}$). On a lattice of spacing h the stochastic Allen–Cahn field has energy and Gibbs law

$$\mathcal{H}_h(\phi) = \sum_i \Big[\frac{(\phi_{i+1} - \phi_i)^2}{2h} + h\,V(\phi_i)\Big], \quad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \qquad p(\phi) \propto e^{-\mathcal{H}_h(\phi)/\Theta}. \tag{12}$$

The spring term is the lattice form of $\int \frac{1}{2}(\partial_x\phi)^2$ and the well term of $\int V$; the gradient of $\mathcal{H}_h$ in one coordinate contains the second difference $(\phi_{i+1} - 2\phi_i + \phi_{i-1})/h$, which becomes $\partial_x^2\phi$ as $h \to 0$, giving $\partial_s\phi = \partial_x^2\phi - V'(\phi) + \sqrt{2\Theta}\eta$. Every factor in (12) couples at most two neighbours, so a snapshot is exactly a Markov chain in the site index: in physical terms, a Gaussian random walk (each step of variance Θh) re-weighted at every site by $e^{-hV/\Theta}$.

Transfer operator and the kernel. Split the site factors evenly between the two bonds that touch them: $p \propto \prod_i \mathsf{T}(\phi_i, \phi_{i+1})$ with $\mathsf{T}(\phi, \phi') = D(\phi)\,G(\phi' - \phi)\,D(\phi')$, $D = e^{-hV/2\Theta}$, $G(u) = e^{-u^2/2\Theta h}$. As an integral operator T is symmetric with positive kernel, so by Perron–Frobenius its largest eigenvalue λ_0 is simple, strictly above $\lambda_1 \geq \lambda_2 \geq \ldots$, with an everywhere positive eigenfunction ψ_0. Inside a long chain each half contributes $\mathsf{T}^r\mathbf{1} \propto \psi_0$, so the stationary marginal is $\pi = \psi_0^2$ ($\int \psi_0^2 = 1$), and normalising the continuation weight

$\mathsf{T}(\phi,\phi')\psi_0(\phi')$ by the eigen-equation $\int \mathsf{T}(\phi,\phi')\psi_0(\phi')\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$ gives the *ground-state transform*

$$K(\phi' \mid \phi) = \frac{\mathsf{T}(\phi,\phi')\,\psi_0(\phi')}{\lambda_0\,\psi_0(\phi)}, \qquad \text{reversible:} \quad \pi(\phi)K(\phi' \mid \phi) = \frac{\psi_0(\phi)\mathsf{T}(\phi,\phi')\psi_0(\phi')}{\lambda_0} \text{ is symmetric.} \tag{13}$$

Inserting $\mathsf{T}^r = \sum_n \lambda_n^r \psi_n \psi_n^\top$ into the two-point function gives $\mathbb{E}[\phi_i\phi_{i+r}] = \sum_{n\geq 1}(\lambda_n/\lambda_0)^r c_n^2$ with $c_n = \int \psi_0\,\phi\,\psi_n$: correlations decay geometrically, dominated by λ_1/λ_0, which plays for the field the role ρ plays for the linear chain, and the correlation length is $\xi = 1/\log(\lambda_0/\lambda_1)$ sites. Because $\lambda_0 > \lambda_1$ strictly at every $\Theta > 0$, ξ is finite: *there is no phase transition in one dimension*. Entropy is the reason: a kink costs E_{kink} but can sit anywhere, so a long stretch always contains some, at a density $\sim e^{-E_{\mathrm{kink}}/\Theta}$.

Kinks and the kink energy (derivation). A static kink solves $\phi'' = V'(\phi)$. Multiplying by ϕ' and integrating (both sides are derivatives, and both vanish far from the kink) gives $\frac{1}{2}\phi'^2 = V(\phi)$, so $\phi' = \sqrt{2V} = (1-\phi^2)/\sqrt{2}$, which integrates to $\phi_{\mathrm{kink}}(x) = \tanh(x/\sqrt{2})$ and $E_{\mathrm{kink}} = \int[\frac{1}{2}\phi'^2 + V]\mathrm{d}x = \int_{-1}^{1}\sqrt{2V(\phi)}\,\mathrm{d}\phi = \frac{1}{\sqrt{2}}\int_{-1}^{1}(1-\phi^2)\mathrm{d}\phi = \frac{2\sqrt{2}}{3} \approx 0.943$. The dilute-kink law $\xi \propto \Theta^{1/2}e^{E_{\mathrm{kink}}/\Theta}$ is the prediction that a fit on $\Theta \in [0.1, 0.3]$ should return this energy; it returned 0.988 (registered prediction: within 2%, falsified), so the law is approached as Θ falls but its constants are not pinned.

The tilted-walk identity (proof). Put $g = \psi_0 D$. Then $K(\phi' \mid \phi) = D(\phi)G(\phi'-\phi)g(\phi')/(\lambda_0\psi_0(\phi))$, and the eigen-equation reads $D(\phi)\int G(\phi'-\phi)g(\phi')\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$, which is the normaliser. With $G = \sqrt{2\pi\Theta h}\,\mathcal{N}(\cdot;\phi,\Theta h)$,

$$K(\phi' \mid \phi) = \frac{\mathcal{N}(\phi';\phi,\Theta h)\,g(\phi')}{Z_g(\phi)}, \qquad Z_g(\phi) = \int \mathcal{N}(\phi';\phi,\Theta h)g(\phi')\mathrm{d}\phi'. \tag{14}$$

A *tilted walk* is a Gaussian step re-weighted by a positive function of the arrival point; Z_g is the column normaliser (the Doob h-transform). A model that parametrises $\log g$ and normalises each column numerically contains the truth *without computing ψ_0 or λ_0*.

Reversibility and the pair energy (proof). For any $\gamma > 0$ and positive g, the tilted walk $K_g(a' \mid a) = \mathcal{N}(a';a,\gamma)g(a')/Z_g(a)$ is reversible with invariant density $\pi_g \propto gZ_g$, because $\pi_g(a)K_g(a' \mid a) \propto g(a)g(a')\mathcal{N}(a';a,\gamma)$ is symmetric (detailed balance). It is the Gibbs chain of the symmetric pair energy

$$W(a,a') = \frac{(a'-a)^2}{2\gamma} + f(a) + f(a'), \qquad f = -\tfrac{1}{2}\log(g/Z_g)\,: \tag{15}$$

with $u = e^{-f} = \sqrt{g/Z_g}$ and $\mathsf{T}_g(a,a') = u(a)\mathcal{N}(a';a,\gamma)u(a')$, the function $\psi = \sqrt{gZ_g}$ satisfies $\mathsf{T}_g\psi = uZ_g = \psi$ (using $u\psi = g$), so it is the Perron eigenfunction with eigenvalue 1, and its ground-state transform $\mathsf{T}_g(a,a')\psi(a')/\psi(a)$ equals K_g; the converse is the tilted-walk identity with $D = e^{-f}$. For the field, $g/Z_g = \sqrt{2\pi\Theta h}\,D^2/\lambda_0$, so $f = -\log D + \mathrm{const} = hV/2\Theta + \mathrm{const}$. *So the directed kernel and the symmetric pair energy are one model class; nothing irreversible can be learned by mistake, and a fitted kernel reads as physics.* In standardised units $a = \phi/s$, $s^2 = \mathrm{Var}_\pi\,\phi$, the step variance is $\gamma = \Theta h/s^2$ and the read-out is $\hat{\Theta} = \hat{\gamma}s^2/h$, $\hat{V}(\phi) = (2\hat{\Theta}/h)\hat{f}(\phi/s) + \mathrm{const}$: the temperature from the bond alone, the potential (barrier, wells) from the site energy.

Why the paper's mixture kernel fails on the field (proof). For a tilted walk, differentiating $Z_g(a) = \int \mathcal{N}(a';a,\gamma)g(a')\mathrm{d}a'$ under the integral gives $Z_g'(a) = \int \frac{a'-a}{\gamma}\,\mathcal{N}(a';a,\gamma)\,g(a')\,\mathrm{d}a' = Z_g(a)\,\mathbb{E}[a'-a \mid a]/\gamma$, so

$$\mathbb{E}[a' - a \mid a] = \gamma\,(\log Z_g)'(a). \tag{16}$$

For the field, $\log Z_g = \log\psi_0 + hV/2\Theta + \mathrm{const}$, so in physical units the expected step is $\Theta h(\log\psi_0)'(\phi) + \frac{h^2}{2}V'(\phi)$: a nonlinear function with zeros inside the wells, pushing sites off the barrier towards the domain values. The mixture kernel has $\mathbb{E}[a' \mid a] = \rho a + \mathrm{const}$, linear in a, whatever the shape of its driving noise. It is misspecified through its *drift*, not through the shape of its noise, so as $M \to \infty$ EM converges to its KL-closest member, the linear-Gaussian projection, and the risk equals the linear floor (1.00–1.02 times it at $M = 2048$): no amount of data repairs it. The two laboratories intersect only at the Gaussian chain: $\mathcal{N}(a';\rho a, q)$ is a tilted walk with $\gamma = q/\rho$ and $\log g(a') = -(1-\rho)a'^2/2q$ (complete the square), and for a chain $k(a'-\rho a)$ the mixed derivative $\partial_a\partial_{a'}\log K = -\rho(\log k)''$ is the constant $1/\gamma$ only if $\log k$ is quadratic.

11 What noise erases: Hermite attenuation and information decay

This section is the supporting theory of the paper (its Section 6.1). It says where structure beyond second order lives and why it fades first. Write He_n for the probabilists' Hermite polynomials and $\alpha = e^{-t}$.

Theorem 3 (Hermite attenuation; proof). For $z \sim \mathcal{N}(0,1)$, $\mathbb{E}_z \mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2} z) = \alpha^n \mathrm{He}_n(a)$. *Proof.* Average the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$ at $x = \alpha a + \sqrt{1-\alpha^2} z$ over z: it gives $e^{s\alpha a - s^2\alpha^2/2} = \sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$, and matching powers of s proves the claim. For multi-indices ν the coordinates of $\mathbf{z}$ are independent, so the product form $\mathbb{E}[H_\nu(\mathbf{x}) \mid \mathbf{a}] = e^{-|\nu| t} H_\nu(\mathbf{a})$ holds. □ So the channel multiplies the Hermite moment of degree m by e^{-mt} exactly. For cumulants: noise only adds to the second cumulant, so the cumulant of order $m \geq 3$ of $\mathbf{x}$ is e^{-mt} times that of $\mathbf{a}$: the correlation structure (degree 2) fades as e^{-2t} and the fourth cumulant κ_4 as e^{-4t}. Diffusion noise is a low-pass filter on statistical order.

Exact identity for the log-density. With $s = \alpha \mathbf{x}/\Delta_t$ and $b = \alpha^2/\Delta_t$, Bayes' rule gives $P_t(\mathbf{x}) = g_{\Delta_t}(\mathbf{x})\, \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2)$, with g_Δ the $\mathcal{N}(0, \Delta I)$ density. Hence $\log P_t(\mathbf{x}) = \log g_{\Delta_t}(\mathbf{x}) + \Lambda(s,b)$, $\Lambda(s,b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2)$, $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \nabla_s \Lambda$ and $S = -\mathbf{x}/\Delta_t + (\alpha/\Delta_t)\nabla_s \Lambda$ (this is Tweedie again). The Taylor coefficients of Λ at the origin are joint cumulants of $s \cdot \mathbf{a}$ and $|\mathbf{a}|^2/2$; this needs an exponential moment ($\mathbb{E} e^{\lambda |\mathbf{a}|} < \infty$, true for all the laws except Student-t).

Theorem 5 (order-by-order decay; sketch). Grade a monomial $s^j b^k$ by $j + 2k$, which is its order in α since $s = O(\alpha)$, $b = O(\alpha^2)$. By the symmetry $\mathbf{a} \mapsto -\mathbf{a}$ only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$. The grade-two part depends on Σ_0 only; the grade-four part differs from that of the covariance twin by a term *linear in the fourth-cumulant tensor* κ_4. Hence the leading term of the score gap is the gradient of the Wick (Hermite) product,

$$S - S^{\mathrm{G}} \approx \alpha^4 \nabla_{\mathbf{x}} \Big[\tfrac{1}{24} \sum_{ijkl} \kappa_4[ijkl] :x_i x_j x_k x_l: \Big], \qquad \text{component } m: \ \alpha^4 \tfrac{1}{6} \sum_{jkl} \kappa_4[mjkl] :x_j x_k x_l:, \tag{17}$$

and by Wick's theorem the second moment of $:x_j x_k x_l:$ against itself is the number of index-respecting perfect matchings, 3! times the symmetric contraction, so $\mathbb{E}\big[(\frac{1}{6}\sum \kappa_4[mjkl] :x_j x_k x_l:)^2\big] = \frac{3!}{36} \sum_{jkl} \kappa_4[mjkl]^2$; summing over m:

$$\mathbb{E}_{P_t} \|S - S^{\mathrm{G}}\|^2 = \tfrac{1}{6} \|\kappa_4\|_F^2 \, e^{-8t} \left(1 + O(e^{-2t})\right). \tag{18}$$

The error from replacing the reference law by P_t is a relative $O(\alpha^2)$, the sixth-grade remainder is $O(\alpha^{10})$ in the cross term, and the tail outside the region where the expansion holds has probability $\exp(-c\, e^{2t})$, below every power of α. Without the exponential-moment assumption (Student-t) the tail is polynomial and the rate is heavy-tailed; this is the regime of the measured t_5 chain, whose local slope on the last interval is -5.28 rather than -8. **Information.** Since $\partial_\theta \log P_t = \partial_\theta \Lambda$, the Fisher information about a parameter θ of the clean law carried by one noised sequence is

$$I_t(\theta) = \begin{cases} \frac{1}{2} \|\partial_\theta \Sigma_0\|_F^2 \, e^{-4t} \, (1 + \epsilon_t) & \partial_\theta \Sigma_0 \neq 0 \quad \text{(covariance coordinates)}, \\ \frac{1}{24} \|\partial_\theta \kappa_4\|_F^2 \, e^{-8t} \, (1 + \epsilon_t) & \partial_\theta \Sigma_0 = 0 \quad \text{(shape coordinates)}, \end{cases} \tag{19}$$

by Wick's theorem applied to the leading grade ($\frac{\alpha^4}{4} \cdot 2\|\partial \Sigma_0\|^2$, respectively $\frac{4!}{24^2} \|\partial \kappa_4\|^2 \alpha^8$). **General order** (Theorem 6): if the lowest non-vanishing Hermite moment of $\partial_\theta P_0$ has degree m, $I_t(\theta) = e^{-2mt} \sum_{|\nu|=m} (\partial_\theta \mathbb{E}[H_\nu(\mathbf{a})])^2/\nu! \cdot (1 + O(e^{-2t}))$ by orthogonality of the H_ν under the reference Gaussian. Degree-2 and degree-4 coordinates are orthogonal at leading order, so the efficient information about a shape coordinate after the correlation is estimated equals its raw information.

Corollary 7 (sample complexity by order). At fixed t the family is differentiable in quadratic mean with finite non-singular information, so the Hájek–Le Cam local asymptotic minimax bound holds and maximum likelihood attains it: any estimator from M sequences at level t has risk at least $1/(M I_t^{\mathrm{eff}}) \asymp e^{2mt}/M$. A covariance coordinate therefore needs $M \propto e^{4t}$ sequences for a fixed precision and a shape coordinate $M \propto e^{8t}$: the ratio e^{4t} is the extra cost of shape. This bound is for regular parametric families at fixed t and says nothing about absolute precision beyond $1/\sqrt{M I_t}$ exceeding the coordinate's range.

The crossover t_*. For the Gaussian chain, P_t / g_{Δ_t} is expanded in $\alpha^2 \Sigma_0$, i.e. in $e^{-2t} \lambda_{\max}(\Sigma_0)$ in operator norm, so the expansion is valid for $t > t_* = \frac{1}{2} \log \lambda_{\max}(\Sigma_0)$ ($t_* = 1.17$ at $\rho = 0.85$, where $\lambda_{\max} \leq (1+\rho)/(1-\rho)$

by the Toeplitz bound). Equivalently $P_t/g_{\Delta_t} \in L^2(g_{\Delta_t})$ iff $2I - \Sigma_t \succ 0$, i.e. $e^{2t} > \lambda_{\max} - 1$. Below t_* the long-wavelength covariance modes are still of order one and the measured rates are shallower (slopes -3 to -6 rather than -4 and -8 at $t \lesssim 1.6$).

Information in cumulant coordinates (the Q10 recomputation). The earlier curves were Fisher information about $\theta = (\rho, q, \beta)$ where β is the shape parameter of a generalised-Gaussian noise law of fixed variance q. Jérôme's point: β is not a moment; resolving it needs all cumulants. The honest coordinates are $\phi = (\rho, \kappa_2, \kappa_4)$ with $\kappa_2 = q$ and $\kappa_4 = \gamma_2(\beta)q^2$, $\gamma_2(\beta) = \Gamma(5/\beta)\Gamma(1/\beta)/\Gamma(3/\beta)^2 - 3$ the excess kurtosis. Information transforms as a tensor, not as a label: with the Jacobian $J = \partial\phi/\partial\theta$ (lower triangular, last row $(0,\ 2\gamma_2 q,\ \gamma_2' q^2)$),

$$I_\phi = J^{-\top} I_\theta J^{-1}, \qquad I^{\text{eff}}(\phi_k) = \frac{1}{(I_\phi^{-1})_{kk}} = \frac{1}{j_k^\top I_\theta^{-1} j_k}, \tag{20}$$

with j_k the kth row of J. For ρ and κ_2 the rows are unit vectors, so the effective information is unchanged; for κ_4 the row mixes in the q direction ($2\gamma_2 q \neq 0$), so $I^{\text{eff}}(\kappa_4) \neq I^{\text{eff}}(\beta)$, by a factor between 1.6 and 1526 across the schedule: a relabel would have been wrong. In these coordinates the ratio of fourth- to second-order information at the Laplace member is 0.071 at $t = 0.05$ and 9.9×10^{-5} at $t = 3$, a fall by a factor of about 700, and the fitted slopes on the smooth members are -3.99 (κ_2) and -7.94, -8.01 (κ_4, $\beta = 2, 3$), against the predicted -4 and -8.

12 Where the structure stops

A hidden variable shared by the whole sequence. Let $\mathbf{X} = Gb\mathbf{1} + \mathbf{U}$, $G = \pm 1$ a fair sign per sequence and $\mathbf{U}$ the Gaussian linear chain. Given G the sequence is Markov; marginally it is not (the whole history informs G). Noised, $\mathbf{x} \mid G{=}g \sim \mathcal{N}(g\mathbf{m}, \Sigma_t)$ with $\mathbf{m} = e^{-t}b\mathbf{1}$ and $\Sigma_t = e^{-2t}\Sigma_U + \Delta_t I$, so $P_t = \frac{1}{2}\sum_g \mathcal{N}(g\mathbf{m}, \Sigma_t)$. *Exact score (proof).* $\nabla \log P_t = \sum_g w_g\,(-\Sigma_t^{-1}(\mathbf{x} - g\mathbf{m}))$ with posterior weights $w_g \propto \exp(g\,\mathbf{m}^\top \Sigma_t^{-1}\mathbf{x})$, so

$$S(\mathbf{x}, t) = -\Sigma_t^{-1}\big(\mathbf{x} - \mathbf{m}\tanh(\mathbf{m}^\top \Sigma_t^{-1}\mathbf{x})\big), \tag{21}$$

available without any grid: the reference. The linear floor uses the true covariance $\Sigma_U + b^2\mathbf{1}\mathbf{1}^\top$. *The bias that data cannot remove.* A first-order kernel fitted by maximum likelihood converges, as $M \to \infty$, to the member of its family closest in KL to the noised law (the maximum-likelihood limit under a misspecified model). For the Gaussian version of the latent problem this is $\theta^* = \arg\min_{(\rho,q)} \sum_t \mathrm{KL}\big(\mathcal{N}(0, \tilde\Sigma_t) \,\|\, \mathcal{N}(0, \hat\Sigma_t(\rho, q))\big)$, and the resulting score $-\hat\Sigma_t^{-1}\mathbf{x}$ differs from the truth $-\tilde\Sigma_t^{-1}\mathbf{x}$ by a pooled relative error

$$b^2 = \frac{\sum_t \operatorname{tr}\left[(\hat\Sigma_t^{-1} - \tilde\Sigma_t^{-1})\tilde\Sigma_t(\hat\Sigma_t^{-1} - \tilde\Sigma_t^{-1})\right]}{\sum_t \operatorname{tr}\tilde\Sigma_t^{-1}} \tag{22}$$

(using $\mathbb{E}\|A\mathbf{x}\|^2 = \operatorname{tr}(A\tilde\Sigma_t A^\top)$ and $\mathbb{E}\|\tilde\Sigma_t^{-1}\mathbf{x}\|^2 = \operatorname{tr}\tilde\Sigma_t^{-1}$). The finite-sample error is $\sqrt{b^2 + v/M}$: a bias b that no sample size removes plus a variance that falls like $1/M$; the measured values reproduce it to within 0.019 across mechanisms. *The repair.* Enlarge the state to (X_i, G): a hidden Markov model whose latent chain never changes, i.e. a mixture of two chains with weights inferred per sequence from the evidences. With the reflection symmetry of the truth, the minus chain is the reflection of the plus chain, $K_-(x' \mid x) = K_+(-x' \mid -x)$, $\mu_-(x) = \mu_+(-x)$, so the augmented model has the parameters of one kernel and none for b (a drift constant already encodes the offset). BP runs on each component separately with its evidence Z_g; the responsibilities are $r_g \propto \pi_g Z_g$; the posterior mean is $\sum_g r_g \mathbb{E}[a_i \mid \mathbf{x}, g]$; and the E-step sums the components' pair statistics weighted by r_g with the minus component reflected back onto the plus frame. Fisher's identity holds for the mixture evidence $Z = \frac{1}{2}(Z_+ + Z_-)$, so EM is exact maximum likelihood for the augmented model. The reflection also explains the surprise at $b = 0$: fitting the plain kernel on each sequence *and its sign-flipped copy* uses the symmetry and equals, in average risk, fitting on twice as many sequences.

Subsampling. Observing every s-th site of a Markov chain gives a Markov chain with kernel $K^s = K \circ \cdots \circ K$ (Chapman–Kolmogorov). In the eigenbasis of the normalised transfer operator its eigenvalues are $(\lambda_n/\lambda_0)^s$, so the correlation length in the new index is ξ/s and the coarse-grained physics is unchanged, but K^s is no

longer a Gaussian step times a function of the arrival point, which is why the 5- and 25-number kernels become misspecified (risk ×25 and ×14 from stride 1 to 4) while a network kernel can follow it.

A chain outside the family. The Laplace-bond chain is built, like the field, from a symmetric pair weight, here $e^{W(a,a')}$ with $W(a,a') = -|a'-a|/\lambda - \eta[V(a)+V(a')]/2$, $V(a) = (a^2-1)^2/4$, $\lambda = 0.40$, $\eta = 2.0$ (chosen before any fit to match the base cell's lag-one correlation and kurtosis), by the same Perron–Frobenius construction on the grid. Its kernel is the ground-state transform of this weight and is not a Gaussian step times a function of the arrival point (a Laplace bond is not Gaussian), so the tilted-walk families are misspecified there; the oracle and exact samples come from the grid, with no discretisation gap between data and reference.

13 Speciation: the commitment of the denoiser

For a noised sequence the exact posterior gives $p_i = P(a_i > 0 \mid \mathbf{x})$ (the sign of a site is the domain it belongs to), and the *commitment* of site i is $|2p_i - 1| \in [0,1]$. Averaged over sites and data it falls from 1 at $t = 0$ to 0 at high noise; t_c is the level where it crosses $\frac{1}{2}$. **Calibration.** If p_i is the true posterior probability that the sign is +, then guessing the sign by the posterior's majority is right with probability $\mathbb{E}[\max(p_i, 1-p_i)] = \frac{1}{2}(1 + \mathbb{E}|2p_i - 1|)$: the commitment is the excess probability of guessing right, which E50 verifies to within 0.003 on average over levels. **The Gaussian twin in closed form (proof).** For the covariance twin the posterior of a_i is $\mathcal{N}(\mu_i, v_i)$ with v_i not depending on the data and μ_i a linear function of $\mathbf{x}$, with $\operatorname{Var}\mu_i = \Sigma_{ii} - v_i$ by the law of total variance. So (a_i, μ_i) is jointly Gaussian with correlation $R_i = \sqrt{1 - v_i/\Sigma_{ii}}$, and the probability that the sign of μ_i agrees with the sign of a_i is the orthant probability $\frac{1}{2} + \frac{1}{\pi}\arcsin R_i$. By calibration the commitment is $2 \cdot P(\text{right}) - 1$:

$$\text{commitment}_i = \tfrac{2}{\pi}\arcsin R_i. \tag{23}$$

It equals $\frac{1}{2}$ exactly when $R_i = \sin(\pi/4)$, i.e. $v_i = \Sigma_{ii}/2$: the twin is half-committed where its posterior variance has halved. For a single Gaussian mode of prior variance λ seen through the channel, the posterior precision is $1/\lambda + e^{-2t}/\Delta_t$, so $v/\lambda = \left(1 + \lambda e^{-2t}/\Delta_t\right)^{-1} = \frac{1}{2}$ iff $\lambda e^{-2t} = \Delta_t = 1 - e^{-2t}$, i.e. $e^{2t} = 1 + \lambda$:

$$t_c = \tfrac{1}{2}\ln(1 + \lambda_{\max}), \tag{24}$$

the commitment time of the leading covariance mode, which for chains shorter than the correlation length matches the measured t_c one-for-one (slope 0.98): the finite-chain analogue of the speciation time $\frac{1}{2}\ln\Lambda$ of Biroli, Bonnaire, de Bortoli and Mézard (2024) for the largest eigenvalue Λ of the data covariance. The exact (non-Gaussian) posterior commits earlier than its twin, at every temperature, by 0.47 in t when coldest and 0.07 when hottest: the two humps carry commitment information that the covariance alone does not.

14 Cost, numerics and the grid

Operation counts. Per sequence and level, grid BP costs $O(LN_g^2)$ (two sweeps plus the pair marginals). Measured log–log slopes on the 8-thread node: 1.03 in the batch, 1.03 in L, and 1.16 in N_g rather than 2, because at these sizes element-wise work and memory traffic, not the matrix product, dominate (the matrix product is only 12–15% of a sweep at $N_g = 401$). **Amdahl bound.** If a fraction f of the time is a step that is made *free*, the speed-up is at most $1/(1-f)$: with $f = 0.12$–0.15 no faster solver for the transition step can exceed 1.14–1.18×. An FFT transition computes the convolution in $O(N_g \log N_g)$ per vector instead of N_g^2; ours is second-order accurate (6.7×10^{-4} from dense BP) and 5.8 times *slower* at $N_g = 401$, and still 2.2 times slower at $N_g = 3201$, the largest grid timed: the asymptotic advantage is not reached at the sizes that matter here. **Fourier representations of messages.** A message is a function; one can store its first D Fourier coefficients on a window instead of N_g grid values. On the Laplace chain a Fourier basis adapted to each message needs 33 numbers for a KL divergence of 10^{-2} and 47 for 10^{-4} against 45 and 151 for the grid, but reaches 10^{-4} in only 53 of 144 cases and adding modes can make the error *increase*, so a solver cannot choose a truncation and know its error. The Gram–Charlier (cumulant) expansion is non-monotone in most cases. **Closure families for messages.** A family is *closed* if the recursion (5) maps it into itself. The Gaussian is (on a Gaussian chain, Proposition 2). The family "a Gaussian passed once through the kernel" (Jérôme's

Laplace-type ansatz) is exact at site 1 only and not closed (432/432 configurations fail in the interior, KL about 1.6×10^{-5} at site 31 at low noise); it is still 2,272 times closer to the true message than a Gaussian. **Coarser grids.** With the true ϕ^4 kernel the integrands are smooth and $N_g = 101$ is exact to 2×10^{-11}; the learned ϕ^4 kernels lose nothing at 101 points and at most 1.2% at 51, and EM runs about 14 times faster. The linear chain's 8-component mixtures learn components narrower than a 101-point spacing, so they need about 151 points; a kernel component narrower than two grid cells is flagged "unresolved". **Memory.** One BP call on 512 sequences of 32 sites at 401 points holds five message-sized arrays, about 260 MB, linear in batch, length and grid.

15 Experimental protocol and statistics

Estimands and data. Risks are (11): on all sites (headline), on interior sites 5–28 (used for selection), and on the centre site (diagnostic). Twelve log-spaced levels $t \in [0.05, 3]$. Per seed, a validation bundle (1024 sequences) and a test bundle (2048) are drawn from streams disjoint from every training stream, and their hashes are stored with every row so pairing across arms is checkable. **Seeds and statistics.** Sixteen confirmatory seeds for every comparison at the frozen configuration (eight for regime sweeps and some ϕ^4 studies); a seed is an independent draw of data and initialisation. Ratios are formed *within seed* (paired by test bundle) and averaged; intervals are percentile bootstraps over seeds; "16 of 16 seeds" counts seeds in which an arm is better. Development seeds, disjoint from the confirmatory ones, are used for every architecture and hyperparameter choice (three restarts per configuration). **Interfaces.** (A) networks trained by DSM on clean sequences with the level and corruption redrawn every step; (B) the same on one fixed corruption per clean sequence; (C) ISM on the same one-view noised bundle as EM–BP, the checkpoint selected on the held-out implicit objective, so the true score enters only in the final scoring. EM–BP is always at interface C. **Registry discipline.** For each experiment a registry file with the hypothesis, the design, the predictions and the analysis script is written *before* the main run, with dated amendments; outputs are frozen by SHA-256 checksum (320 files at the last check, 0 problems); a prediction is read by its own rule as CONFIRMED, FALSIFIED, or UNRESOLVED when the rule cannot decide (for example a fit still improving when its budget ends); a failed fit counts as a failure and is never dropped. **From files to the manuscript.** Every number in the paper is a macro generated from the frozen files; a check regenerates the fragments in a scratch tree and diffs them, rejects any typed decimal, and builds the paper; an independent script recomputes headline numbers with plain pandas and compares.

Sources: `experiments/registry/E*.json` (hypotheses, predictions, outcomes, corrections) and the frozen outputs they list; `reports/field-laboratory-2026-10-01.pdf` (full detail); `docs/JEROME_CALL_2026-09-25.md` (the question ledger). Branch `claude/phi4-field-2026-10-01`.